

PREDICTING CUSTOMER CHURN: A DATA-DRIVEN ANALYSIS OF SUBSCRIPTION CUSTOMER BEHAVIOUR USING CLASSIFICATION MODELS

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Data Science and AI

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India Data Research Academy

DECLARATION

I hereby declare that the project report titled "Predicting Customer Churn: A Data-Driven Analysis of Subscription Customer Behaviour Using Classification Models", submitted in fulfillment of the requirements of the Data Science and AI Summer Training Program conducted by the India Data Research Academy (IDRA), is a record of original work carried out by me under the supervision of Dr. Shaheena Salam.

I further declare that the data cleaning, exploratory analysis, statistical testing, feature engineering, model development and evaluation presented in this report were performed by me, and that the accompanying Google Colab notebook reproduces every result reported here. All external sources, datasets, software libraries and reference materials used in the course of this work have been duly acknowledged in the References section.

No part of this report has been copied from another student, published work or online source without appropriate citation, and this work has not been submitted previously for the award of any other certificate, diploma or degree.

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I am especially grateful to my supervisor, Dr. Shaheena Salam, whose guidance on framing an analytical question, justifying methodological decisions and interpreting results in their business context shaped the direction of this work considerably. The emphasis placed throughout the programme on explaining why a decision was made, rather than simply performing an operation, is reflected in every section of this report.

I acknowledge the IDRA Learning Management System for providing the Telco Customer Churn dataset used in this study, and the open-source community behind Python, Pandas, NumPy, Matplotlib, Seaborn, SciPy and Scikit-learn, whose libraries made this analysis possible. Finally, I thank my institute, my faculty members and my peers for their continued encouragement and constructive feedback during the twenty-day project window.

ABSTRACT

Customer churn is one of the most consequential problems in subscription businesses, where retaining an existing customer is substantially cheaper than acquiring a replacement. This study analyses a telecommunications subscriber dataset of 7,043 customers described by 20 attributes covering demographics, contracted services, tenure and billing behaviour, with the objective of predicting whether a customer will churn and identifying the factors that drive that decision. The dataset was assessed for data quality issues and three were addressed: eleven blank entries in TotalCharges were identified as structurally zero values belonging to customers with zero months of tenure and imputed accordingly rather than deleted; seven service columns containing the redundant labels "No internet service" and "No phone service" were consolidated to "No"; and SeniorCitizen was recast from a binary integer to a categorical variable. No observations were discarded. Three features were engineered — banded tenure, a count of subscribed value-added services, and a ratio of current to historical average monthly spend. Exploratory analysis and chi-square, Welch t and Mann-Whitney tests established that contract type, tenure, internet service type and payment method are significantly associated with churn, while gender is not. Five classifiers were compared using stratified five-fold cross-validation. A class-weighted logistic regression was selected as the final model, achieving 73.31% accuracy, 0.4983 precision, 0.7914 recall, 0.6116 F1-score and 0.8423 ROC-AUC on a held-out test set of 1,409 customers, with a training-to-test AUC gap of only 0.008. Recall was prioritised over accuracy because the cost of failing to identify a departing customer greatly exceeds the cost of an unnecessary retention offer. Month-to-month contracts, first-year tenure, fibre-optic service and electronic-check payment emerged as the dominant risk indicators.

Keywords: *Customer churn; Binary classification; Logistic regression; Exploratory data analysis; Data preprocessing; Predictive analytics; Class imbalance; Retention modelling*

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1. INTRODUCTION

1.1 Background

Subscription-based businesses — telecommunications operators, streaming platforms, software-as-a-service vendors and utility providers — derive their revenue from recurring relationships rather than one-off transactions. In such a business the single most damaging event is not a lost sale but a lost subscriber, because the loss recurs every billing cycle thereafter. This event is known as churn: the decision by a customer to terminate their service.

The commercial arithmetic is well established. Acquiring a new customer typically costs several times more than retaining an existing one, and an existing customer has already absorbed their acquisition cost while continuing to generate margin. A mature operator with a stable subscriber base can therefore improve profitability more reliably by reducing churn by a few percentage points than by an equivalent effort spent on acquisition. Churn is also a leading indicator of underlying problems — price dissatisfaction, service quality failures or a competitor advantage — that would otherwise surface only in aggregate revenue figures, by which point the cause has become difficult to isolate.

The difficulty is that churn is usually observed only after it has happened. By the time a cancellation request is received, the customer has already evaluated alternatives and made a decision; intervention at that stage is expensive and frequently unsuccessful. The value of predictive analytics lies precisely in shifting this intervention window earlier, using behavioural and contractual signals that are observable while the customer is still active.

1.2 Context of the Problem

This study uses a telecommunications subscriber dataset in which each row represents one customer of a fictional operator, described by the services they have contracted, the length of their relationship with the company, their billing arrangements, a small set of demographic attributes, and a flag indicating whether they left during the most recent month. The setting is a realistic representation of the customer database any operator maintains.

Several stakeholders would use the output of such an analysis. A retention team requires a ranked list of at-risk customers so that limited outreach capacity is directed where it will matter, rather than being spread uniformly across a subscriber base in which roughly three quarters of customers were never going to leave. A product manager requires the drivers of churn, not merely the predictions, in order to decide whether a service line needs repricing or a support process needs redesigning. A finance function requires an estimate of expected revenue attrition for forecasting. Each of these needs is served by a model that is both accurate and interpretable — a consideration that directly shaped the model selection in Section 11.

1.3 Scope of the Study

This project covers the complete analytical pipeline applied to a single cross-sectional dataset: data quality assessment and cleaning, feature engineering, exploratory data analysis, formal statistical testing of the associations observed, preprocessing with leakage control, comparative training of five classification algorithms, evaluation on a held-out test set, examination of overfitting, decision-threshold analysis and interpretation of feature importance.

The study does not cover the following, and its conclusions should be read with these boundaries in mind. It makes no causal claims: the dataset is observational and contains no experimental variation, so every relationship reported is an association. It does not model the timing of churn, only its occurrence, because the data contains no timestamps and represents a single snapshot rather than a longitudinal record. It does not perform customer lifetime value modelling or a formal cost-benefit optimisation of retention spending, although the threshold analysis in Section 12.4 provides the input such an exercise would require. Finally, it does not extend to deep learning or exhaustive automated hyperparameter search, in keeping with the guidance that techniques should be appropriate to the problem and aligned with the methods covered in the training programme.

2. PROBLEM STATEMENT AND MOTIVATION

2.1 Problem Statement

The central problem this project addresses is stated as follows:

Can a subscription customer's likelihood of churning be predicted from their contract, service and billing attributes while they are still active, and which of those attributes most strongly distinguish customers who leave from those who stay?

This decomposes into a predictive task and an explanatory task. The predictive task is a binary classification problem: assign each customer to the class "will churn" or "will not churn". The explanatory task is to identify and quantify the attributes that carry the predictive signal, so that the organisation can act on causes rather than merely react to predictions.

2.2 Motivation

Three considerations motivate a data-driven solution over the alternatives currently in common use.

- Untargeted retention is wasteful. In this dataset 73.5% of customers did not churn. A blanket retention campaign spends roughly three quarters of its budget on customers who were never at risk, and discounts offered to them are pure margin erosion.*
- Intuition-based targeting is unreliable. Plausible heuristics frequently fail: this analysis shows that gender, an attribute often segmented on, has no statistically significant association with churn ($p = 0.487$), while fibre-optic subscribers — the premium, highest-revenue segment — churn at more than twice the overall rate.*
- The cost of the two error types is asymmetric. Failing to identify a customer who leaves forfeits their entire remaining lifetime value; incorrectly flagging a loyal customer costs one retention offer. Any solution must be tuned to this asymmetry, which is a modelling decision that cannot be made without quantitative analysis.*

A model that identifies a substantial majority of departing customers in advance, and explains what distinguishes them, converts retention from a reactive expense into a targeted and measurable programme.

3. OBJECTIVES

The project was carried out against the following six measurable objectives.

- 1. To assess and document the quality of the customer churn dataset, resolving missing values, inconsistent category labels, incorrect data types and any invalid observations, and to demonstrate through validation that each issue was resolved.*
- 2. To quantify the overall churn rate and profile the customer segments in which churn is concentrated, using descriptive statistics and targeted visualisation.*
- 3. To determine, using formal statistical tests rather than visual inspection alone, which contractual, service and billing variables are significantly associated with churn, and to report the strength of each association through an appropriate effect-size measure.*
- 4. To engineer features that capture churn behaviour more effectively than the raw variables, and to justify each on domain grounds.*
- 5. To develop and comparatively evaluate a set of classification models that predict customer churn, selecting a final model on criteria appropriate to the business cost structure rather than on accuracy alone.*
- 6. To evaluate the selected model using accuracy, precision, recall, F1-score, ROC-AUC and a confusion matrix, to test explicitly for overfitting, and to translate the results into specific, evidence-based retention recommendations.*

4. RESEARCH QUESTIONS

The analysis was structured around five research questions. The section in which each is answered is indicated.

#	Research Question	Answered in
RQ1	What are the principal data quality issues in the dataset, and how should each be treated without introducing bias or losing valid observations?	Section 6
RQ2	What is the overall churn rate, and is the target sufficiently imbalanced to affect the choice of evaluation metric?	Sections 5.4, 8.1
RQ3	Which customer attributes are most strongly associated with churn, and are those associations statistically significant or artefacts of sample size?	Sections 8, 9
RQ4	How accurately can churn be predicted from the available attributes, and which classification algorithm performs best given the business cost of each error type?	Sections 11, 12
RQ5	Does the selected model generalise to unseen customers, or does its performance depend on having memorised the training data?	Section 12.3

Table 1. Research questions and the sections in which they are addressed

These questions guided the analysis rather than describing the code that was run. Each is returned to explicitly in Section 13 (Results and Findings).

5. DATASET DESCRIPTION AND DATA UNDERSTANDING

5.1 Data Source

The dataset used in this study is the Telco Customer Churn dataset, provided through the IDRA Learning Management System as the designated dataset for Capstone Project Title 4, "Predicting Customer Churn: A Data-Driven Analysis of Subscription Customer Behaviour". The file was supplied in comma-separated value format under the name Customer-Churn_.csv and was accessed in September 2026.

The dataset originates from a widely used IBM sample telecommunications dataset and describes a fictional operator providing home phone and internet services. It is distributed for educational and analytical purposes and contains no personally identifiable information: the customerID field is a synthetic alphanumeric key. Its use here is consistent with the licensing and usage terms under which it was provided.

5.2 Dataset Size and Structure

The dataset contains 7,043 rows and 21 columns. Each row is a distinct customer, verified by the fact that the 7,043 customerID values are entirely unique. The target variable is Churn, a binary categorical field recording whether the customer terminated service during the most recent month.

Property	Value
File name	Customer-Churn_.csv
Number of observations (rows)	7,043 customers
Number of variables (columns)	21
Numerical variables (as supplied)	2 (tenure, MonthlyCharges)
Categorical variables (as supplied)	18
Mis-typed variable	1 (TotalCharges, supplied as text)
Identifier column	customerID (7,043 unique values)
Target variable	Churn (Yes / No)
Target distribution	No: 5,174 (73.46%) Yes: 1,869 (26.54%)
Declared missing values (NaN)	0
Hidden missing values	11 (blank strings in TotalCharges)
Duplicate rows	0
Unit of observation	One active or recently departed customer
Observation period	Single cross-sectional snapshot (no timestamps)

Table 2. Summary of the dataset

As Table 2 records, the dataset is well populated and structurally clean: there are no duplicate records and no explicitly declared missing values. The two issues requiring attention are the incorrect data type on TotalCharges, which conceals eleven blank entries, and the absence of any time dimension, which constrains the analysis to cross-sectional rather than survival modelling.

5.3 Data Dictionary

Table 3 documents each variable, its meaning, its type and the role it plays in the analysis. Establishing the feature/target separation at this stage — before any modelling — is what prevents the target or an identifier from inadvertently entering the feature set later.

Variable	Description	Type	Role
customerID	Unique synthetic customer identifier	Categorical (ID)	Excluded
gender	Customer gender (Male / Female)	Categorical (nominal)	Feature

Variable	Description	Type	Role
SeniorCitizen	Whether the customer is a senior citizen	Binary (0/1 → No/Yes)	Feature
Partner	Whether the customer has a partner	Categorical (binary)	Feature
Dependents	Whether the customer has dependents	Categorical (binary)	Feature
tenure	Number of months with the company (0–72)	Numerical (discrete)	Feature
PhoneService	Subscribed to phone service	Categorical (binary)	Feature
MultipleLines	Has multiple telephone lines	Categorical (nominal)	Feature
InternetService	Internet type: DSL / Fiber optic / No	Categorical (nominal)	Feature
OnlineSecurity	Online security add-on subscribed	Categorical (nominal)	Feature
OnlineBackup	Online backup add-on subscribed	Categorical (nominal)	Feature
DeviceProtection	Device protection add-on subscribed	Categorical (nominal)	Feature
TechSupport	Premium technical support subscribed	Categorical (nominal)	Feature
StreamingTV	Streaming television add-on	Categorical (nominal)	Feature
StreamingMovies	Streaming movies add-on	Categorical (nominal)	Feature
Contract	Month-to-month / One year / Two year	Categorical (ordinal)	Feature
PaperlessBilling	Paperless billing enabled	Categorical (binary)	Feature
PaymentMethod	Payment instrument used (4 levels)	Categorical (nominal)	Feature
MonthlyCharges	Current monthly bill amount	Numerical (continuous)	Feature
TotalCharges	Cumulative amount billed to date	Numerical (continuous)	Feature
Churn	Customer left in the last month (Yes / No)	Binary categorical	TARGET

Table 3. Data dictionary for the customer churn dataset

5.4 The Target Variable

Churn is the target because it records the outcome the business needs to anticipate, it is observed for every customer in the dataset, and it is a categorical variable with two mutually exclusive levels — which makes the task binary classification rather than regression.

Of the 7,043 customers, 1,869 (26.54%) churned and 5,174 (73.46%) did not, a ratio of approximately 2.77 to 1. This imbalance is moderate rather than extreme, but it is material to the evaluation strategy. A trivial classifier that predicts "no churn" for every customer would achieve 73.46% accuracy while identifying none of the customers the business cares about. Accuracy is therefore an unsuitable primary metric for this project, and recall on the churn class together with ROC-AUC are used instead. This decision, taken here at the data understanding stage, is carried through Sections 11 and 12.

6. DATA CLEANING

This section documents the condition of the raw data, the treatment applied to each issue identified, and the validation confirming that the issue was resolved. The governing principle throughout was that no observation should be discarded without a reason that can be stated and defended, since dropping rows to eliminate an inconvenience is itself a source of bias.

6.1 Missing Values

A conventional null check returned zero missing values across all 21 columns. This result was not accepted at face value, because missing data frequently hides inside a column that has been loaded with the wrong data type. Inspection of the column types revealed that TotalCharges, which is conceptually a currency amount, had been loaded as text. That is a reliable indicator of non-numeric content.

Filtering the column for whitespace-only entries identified eleven such records — 0.16% of the dataset. These are missing values that a null check cannot detect.

```
# Hidden missing values: detect, then diagnose before treating
blank_mask = df['TotalCharges'].astype(str).str.strip() == ''
print(f'Blank TotalCharges entries: {blank_mask.sum()}')      # -> 11
print(df.loc[blank_mask, 'tenure'].unique())                  # -> [0]

# All eleven belong to customers with tenure = 0: the value is structurally
# zero, not unknown. Convert the column and set those entries to 0.
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
df.loc[df['TotalCharges'].isna(), 'TotalCharges'] = 0.0
```

The diagnostic step is what determines the correct treatment. Every one of the eleven affected customers has a tenure of zero months: they had joined but had not yet completed a billing cycle when the snapshot was taken. Their total charges are not unknown — they are genuinely zero.

Three treatments were available. Deleting the rows would have removed eleven legitimate new customers, precisely the segment in which churn risk is highest, and would have biased the dataset against the newest cohort. Imputing the column median (1,394.55) would have attributed a substantial billing history to customers who have never been billed, contradicting their tenure value and creating an internally inconsistent record. Imputing zero states the fact correctly and preserves consistency with tenure. Zero imputation was therefore applied. Validation after treatment confirmed zero remaining nulls, a float64 data type and a column minimum of 0.0.

6.2 Duplicate Records

Duplicates were checked at two levels. A full-row duplication check returned zero matches, and a check on the customerID key returned zero repeated values: all 7,043 identifiers are unique. No de-duplication was therefore required and the dataset size was unchanged by this step.

The check on customerID is the more informative of the two, since two genuinely distinct records could differ in a single field while still representing the same customer counted twice. Its result confirms that each row is one customer and that no customer is over-represented in the analysis.

6.3 Data Types and Inconsistent Values

Two data type corrections and one category standardisation were applied.

Type corrections

- *TotalCharges was converted from text to float64, as described in Section 6.1.*
- *SeniorCitizen was supplied as an integer taking the values 0 and 1. Although numerically valid, it is conceptually a binary category, not a quantity. Leaving it as an integer would have caused it to be passed to the numerical scaler, where standardising a 0/1 indicator produces a meaningless transformation. It was mapped to "No" and "Yes" so that it is treated as categorical and one-hot encoded consistently with the other binary attributes.*

Inconsistent category labels

Seven service columns — *MultipleLines*, *OnlineSecurity*, *OnlineBackup*, *DeviceProtection*, *TechSupport*, *StreamingTV* and *StreamingMovies* — each carried three levels where two are sufficient. Alongside "Yes" and "No", they contained "No internet service" or "No phone service".

These third levels are redundant. For the analytical question "does this customer have this add-on?", a customer without internet and a customer with internet who declined the add-on both answer "No". Retaining the distinction would have created seven additional dummy columns encoding information already held by *InternetService* and *PhoneService*, inflating dimensionality and introducing multicollinearity into the logistic regression for no analytical gain.

```
# Consolidate redundant category labels across the seven service columns
service_cols = ['MultipleLines', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
               'TechSupport', 'StreamingTV', 'StreamingMovies']

for col in service_cols:
    df[col] = df[col].replace({'No internet service': 'No', 'No phone service': 'No'})

df['SeniorCitizen'] = df['SeniorCitizen'].map({0: 'No', 1: 'Yes'})
```

All seven columns were verified after treatment to contain exactly the two levels "Yes" and "No". A separate check for leading or trailing whitespace across every text column returned no affected columns, and no inconsistent capitalisation was present.

6.4 Outlier Handling

The three numerical variables — *tenure*, *MonthlyCharges* and *TotalCharges* — were assessed using the interquartile range rule, flagging any observation lying more than 1.5 IQR below the first quartile or above the third.

Variable	Q1	Q3	IQR	Lower bound	Upper bound	Outliers	% of data
tenure	9.00	55.00	46.00	-60.00	124.00	0	0.00%
MonthlyCharges	35.50	89.85	54.35	-46.02	171.38	0	0.00%
TotalCharges	398.55	3786.60	3388.05	-4683.52	8868.67	0	0.00%

Table 4. Interquartile range outlier assessment for the numerical variables

Figure 1. Distribution and outlier check for numerical variables

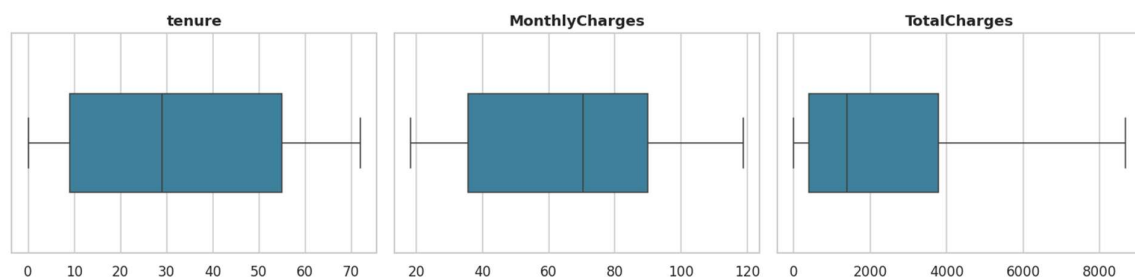


Figure 1. Distribution and outlier assessment of the numerical variables

No outliers were detected in any of the three variables, and consequently none were removed or capped. The boxplots in Figure 1 confirm this visually: no points fall beyond the whiskers in any panel.

TotalCharges is nevertheless visibly right-skewed, with a skewness coefficient of 0.96, a mean of 2,279.73 and a median of 1,394.55. This skew is structural rather than anomalous: total charges are approximately the product of tenure and monthly charge, so customers with long tenures legitimately accumulate large totals. Treating them as errors and removing them would have deleted precisely the loyal, high-value segment that the model needs to recognise as low-risk. The skew was instead addressed at the preprocessing stage through standardisation, which is the appropriate response to scale differences that are real rather than erroneous.

This illustrates a point emphasised in the project guide: a statistical rule flags candidates for inspection, it does not issue instructions. The decision to retain requires domain reasoning about what the values represent.

6.5 Before and After Cleaning

Table 5 summarises the condition of the dataset before and after the cleaning stage and constitutes the validation that each identified issue was resolved.

Data quality check	Before cleaning	After cleaning	Action taken
Rows	7,043	7,043	None — no observations discarded
Columns	21	21	None at this stage
Declared missing values (NaN)	0	0	Not applicable
Hidden missing values	11 (blank strings)	0	Structural-zero imputation
Duplicate rows	0	0	None required
Duplicate customer IDs	0	0	None required
Columns with incorrect type	2	0	TotalCharges → float; SeniorCitizen → categorical
Columns with redundant labels	7	0	Consolidated to Yes / No
Invalid or impossible values	0	0	Verified across five logical checks
IQR outliers (3 numeric columns)	0	0	None detected; none removed

Table 5. Data quality before and after cleaning

The dataset retained its full complement of 7,043 customers through cleaning. This is a deliberate outcome: the only issue that could plausibly have justified deletion — the eleven blank TotalCharges entries — was resolved by correcting the value instead, because diagnosis showed the blanks to be meaningful rather than corrupt.

7. DATA PREPROCESSING

Preprocessing transforms the cleaned dataset into the numerical matrix that a Scikit-learn estimator requires. The critical methodological requirement at this stage is leakage control: any transformation that learns parameters from the data — a scaler learning means and standard deviations, an encoder learning category levels — must learn them from the training partition alone. If a scaler is fitted to the full dataset before splitting, information from the test set influences the training process and the resulting performance estimate is optimistically biased.

This project enforces that requirement structurally rather than by procedural discipline. All transformations are declared inside a Scikit-learn `ColumnTransformer`, which is placed inside a `Pipeline` together with the estimator. The pipeline is then fitted only to the training data, and during cross-validation it is refitted independently on each training fold. No transformation ever observes data it should not.

7.1 Categorical Encoding

After feature engineering (Section 10), 17 categorical features required conversion to numerical form. One-hot encoding was selected over label or ordinal encoding for the following reason: label encoding assigns integers to categories, which implies an ordering and a spacing between them. Encoding `PaymentMethod` as 0, 1, 2 and 3 would assert that credit card payment is "greater than" bank transfer and that the distance between mailed check and electronic check equals the distance between the other pair. No such relationship exists among nominal categories, and a linear model would interpret the fabricated ordering as real.

One-hot encoding creates a separate binary indicator for each level, which carries no ordering assumption. Two configuration choices were made:

- `drop='first'` removes one level from each feature, which becomes the reference category. This avoids the dummy variable trap, in which the full set of indicators for a feature sums to a constant and produces perfect multicollinearity — a condition that makes logistic regression coefficients unstable.
- `handle_unknown='ignore'` ensures that if a category never seen in training appears at prediction time, the pipeline encodes it as all zeros rather than raising an error. This is a robustness measure for any future deployment of the model.

`Contract` is ordinal in principle (month-to-month < one year < two year). It was nevertheless one-hot encoded, because doing so allows the model to learn an independent effect for each level rather than forcing the relationship to be linear in contract length. The results in Section 13 justify this: the effect is markedly non-linear, with the step from month-to-month to one year producing a much larger change in churn odds than the step from one year to two.

7.2 Feature Scaling and Transformation

The six numerical features — `tenure`, `MonthlyCharges`, `TotalCharges`, `NumServices`, `AvgMonthlySpend` and `ChargeRatio` — were standardised using `StandardScaler`, which centres each feature at zero mean and scales it to unit variance.

Scaling is necessary here because the features occupy radically different ranges: `TotalCharges` spans 0 to 8,684.80 while `NumServices` spans 0 to 6. Logistic regression optimises a coefficient per feature, and without scaling the coefficient magnitudes reflect the measurement units rather than the strength of the relationship, making them uninterpretable for comparison. Unscaled features with large ranges also cause the gradient-based solver to converge slowly or not at all within the iteration limit. Standardisation additionally moderates the effect of the right skew in `TotalCharges` without discarding or distorting any observation.

Standardisation is not required by the tree-based models, which split on thresholds and are invariant to monotonic rescaling. It was nonetheless applied uniformly through the shared pipeline because it is harmless to those models and keeps the comparison across the five algorithms strictly like-for-like.

7.3 Train-Test Split

The data was partitioned into 5,634 training observations (80%) and 1,409 test observations (20%), using `random_state = 42` to make the split reproducible.

The split is necessary because a model evaluated on the data it was trained on measures memorisation, not predictive ability. Only performance on observations the model has never encountered estimates how it will behave on genuinely new customers, which is the only performance that matters operationally.

The split was stratified on the target. Random splitting of an imbalanced dataset can by chance produce partitions with differing class proportions, which would make training and test metrics incomparable. Stratification preserves the churn rate in both partitions: 26.54% in training and 26.54% in test, against 26.54% overall.

```
# Stratified split executed BEFORE any transformation is fitted
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y)

# Transformations declared inside the pipeline, never fitted on full data
preprocessor = ColumnTransformer(transformers=[
    ('num', StandardScaler(), numeric_features),
    ('cat', OneHotEncoder(drop='first', handle_unknown='ignore'), categorical_features)])

pipe = Pipeline([('preprocessor', preprocessor), ('classifier', estimator)])
pipe.fit(X_train, y_train) # scaler/encoder learn from training data only
```

7.4 Final Feature Set

Three columns were excluded from the feature matrix. `customerID` is an identifier with 7,043 unique values carrying no generalisable signal; including it would invite the model to memorise individual customers. `Churn` is the target itself, and `Churn_flag` is its numeric encoding — either would constitute direct target leakage.

A leakage audit was conducted over the remaining columns. None records information that would only become available after the churn decision: there is no cancellation date, exit survey, final-bill flag or retention-contact record. Every retained feature describes the customer's state while still active, which is the condition for the model to be usable prospectively.

Group	Count	Variables
Numerical features (scaled)	6	tenure, MonthlyCharges, TotalCharges, NumServices, AvgMonthlySpend, ChargeRatio
Categorical features (one-hot)	17	gender, SeniorCitizen, Partner, Dependents, PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, PaperlessBilling, PaymentMethod, TenureGroup
Total input features	23	Expanding to 29 columns after one-hot encoding
Target	1	Churn (encoded 1 = churned, 0 = retained)
Excluded	1	customerID (identifier, no predictive value)

Table 6. Final feature set used for modelling

8. EXPLORATORY DATA ANALYSIS

The exploratory analysis was directed by the research questions set out in Section 4. Charts were produced to answer a specific question and are presented here only where they support a finding that informs the modelling or the recommendations; supplementary figures are placed in Appendix B. Every figure is followed by an interpretation of what it shows and what decision it supports.

8.1 Univariate Analysis

8.1.1 The target variable

Figure 2. Churn distribution in the customer base

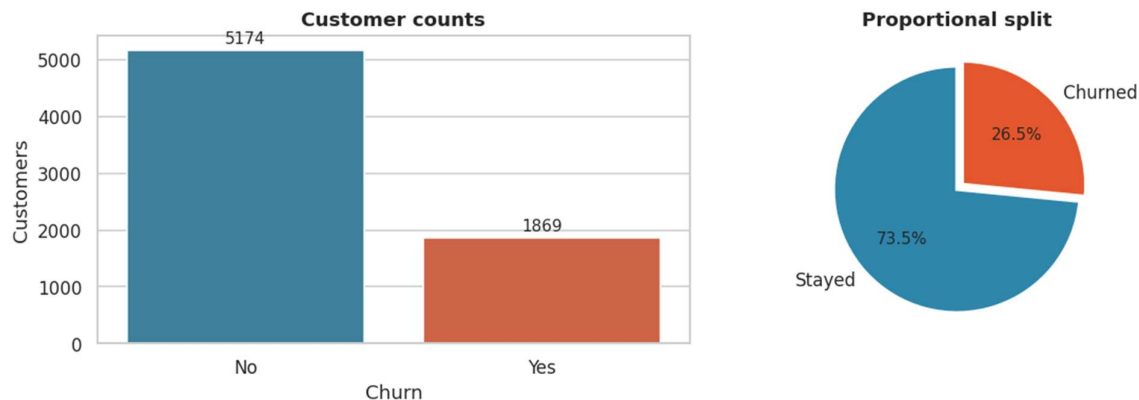


Figure 2. Distribution of the target variable: churned versus retained customers

Figure 2 shows that 1,869 of 7,043 customers (26.54%) churned during the observation month, against 5,174 (73.46%) who did not. Roughly one customer in four left, which is commercially significant: at a mean monthly charge of 74.44 among churners, this represents an annualised revenue exposure in excess of 1.6 million currency units.

Analytically, the distribution confirms that the target is imbalanced at approximately 2.77 to 1. This is the evidential basis for the decision, applied throughout Sections 11 and 12, to treat accuracy as insufficient and to evaluate primarily on recall and ROC-AUC. It also motivates the use of `class_weight='balanced'` in model configuration.

8.1.2 Tenure

Figure 4. Churn risk falls sharply with customer tenure

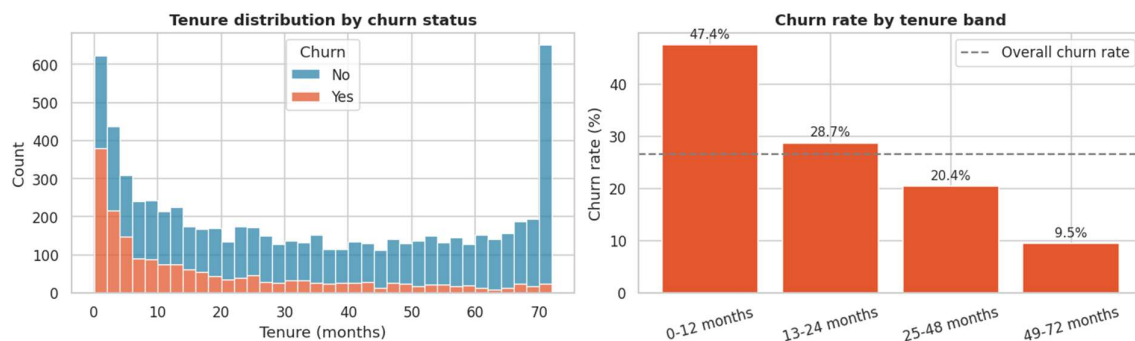


Figure 3. Tenure distribution by churn status and churn rate across tenure bands

The left panel of Figure 3 shows that tenure is bimodal rather than normally distributed. Large clusters occur at both ends of the range: customers in their first few months, and customers approaching the 72-month maximum. The centre of the distribution is comparatively sparse. This is why neither the mean (32.37 months) nor the median (29.00) adequately summarises tenure, and it is the direct justification for the banded `TenureGroup` feature described in Section 10.

The right panel quantifies the lifecycle effect. Churn falls monotonically and steeply with tenure: 47.44% in the first twelve months, 28.71% in months 13 to 24, 20.39% in months 25 to 48 and 9.51% beyond 48 months. Median tenure among churners is 10 months, against 38 months among retained customers. Almost half of all new customers leave within their first year, while customers who pass the four-year mark are five times more stable. The first year is unambiguously the critical retention window, and this finding drives the first recommendation in Section 17.

8.2 Categorical Analysis

8.2.1 Contract type

Figure 3. Contract type is the strongest single discriminator of churn

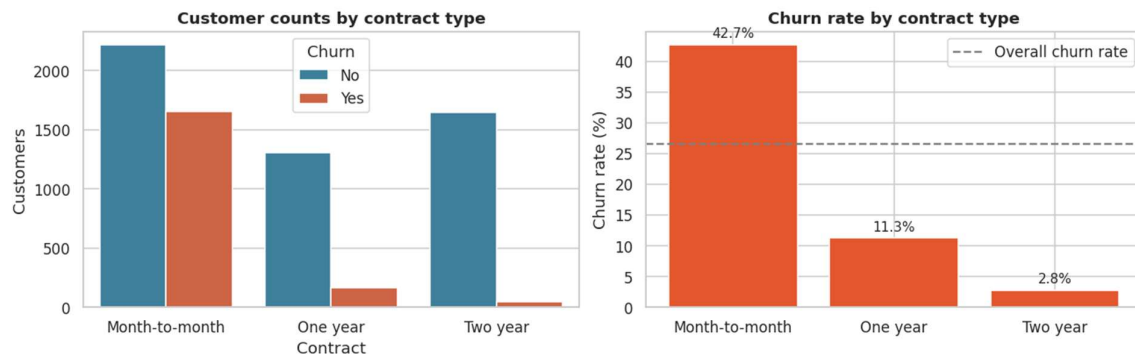


Figure 4. Customer counts and churn rate by contract type

Figure 4 presents the strongest single relationship in the dataset. Month-to-month customers churn at 42.71%; one-year contract holders at 11.27%; two-year contract holders at just 2.83%. The gap between the extremes is a factor of fifteen.

The left panel adds a commercial dimension: month-to-month is also the largest segment, so the highest-risk group is simultaneously the most populous. The interpretation is intuitive — a month-to-month customer faces no switching cost and re-evaluates the relationship every thirty days, whereas a contracted customer faces a penalty and, more importantly, is not prompted to reconsider. Contract type is retained as a core predictor and is confirmed in Section 13 as the feature with the largest protective coefficient.

8.2.2 Internet service and support add-ons

Figure 5. Fibre-optic customers and those without support add-ons churn most

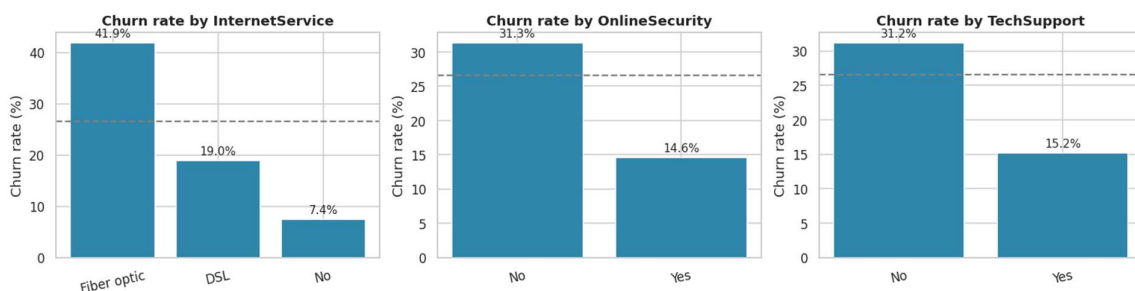


Figure 5. Churn rate by internet service type, online security and technical support

Figure 5 contains the most commercially interesting finding in the analysis. Fibre-optic subscribers churn at 41.9%, against 19.0% for DSL and 7.4% for customers with no internet service at all. The premium, highest-revenue product has by a wide margin the worst retention. Several explanations are consistent with the data — fibre carries a higher price and therefore greater sensitivity to perceived value, it may be the product where service disruptions are most keenly felt, or it may simply attract customers who shop actively on price. The dataset contains no service-quality or competitor-pricing variables and so cannot distinguish between them; this is recorded as a limitation and as a recommendation for investigation rather than presented as a diagnosis.

The two support add-ons show a consistent protective pattern. Customers without OnlineSecurity churn at 31.3% compared with 14.6% for those who have it, and TechSupport shows an almost identical contrast at 31.2% against 15.2% — roughly double the rate in both cases. Two mechanisms are plausible: the add-ons create switching friction and tangible value, or customers who purchase them are inherently more committed. Both are consistent with the association, and this observational dataset cannot separate them. Either way, both variables carry genuine predictive signal and are retained.

8.3 Bivariate and Relationship Analysis

Figure 6. Higher bills and electronic-check payment are associated with churn

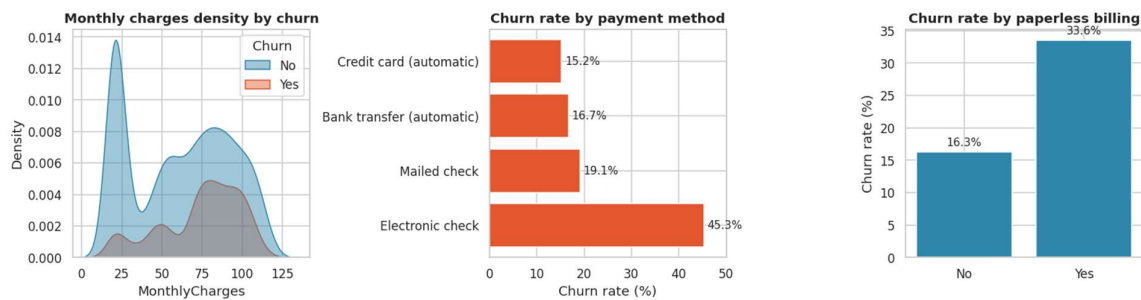


Figure 6. Monthly charges density, payment method and paperless billing against churn

The density plot in the left panel of Figure 6 shows that churned customers are concentrated in the higher monthly-charge range, roughly 70 to 105, while retained customers show a pronounced mass at the low end near 20. Mean monthly charges are 74.44 among churners against 61.27 among retained customers, a difference of 13.17 units. The relationship is positive but the distributions overlap substantially, which tells us that monthly charge alone is a weak discriminator — consistent with its modest correlation of +0.19 reported in Section 8.4, and a useful illustration that a statistically significant difference need not be a practically separable one.

The payment method panel reveals a sharper effect. Customers paying by electronic check churn at 45.3%, more than double the rate of those on automatic bank transfer (16.7%) or credit card (15.2%), with mailed check intermediate at 19.1%. The association is plausibly one of friction and commitment: an automatic payment requires no monthly action and no monthly decision, whereas a manual payment creates a recurring moment at which the customer consciously evaluates whether the service is worth paying for. Paperless billing shows the same direction, 33.6% against 16.3%, though the effect is weaker and partly confounded with the fibre-optic and month-to-month segments.

It should be stressed that these are associations. Migrating a customer to automatic payment will not by itself convert them into a loyal customer if the underlying driver is that less committed customers select manual payment. The recommendation in Section 17 is framed accordingly, as a hypothesis worth testing rather than a guaranteed remedy.

8.4 Correlation Analysis

Figure 8. Pearson correlation matrix of numerical variables

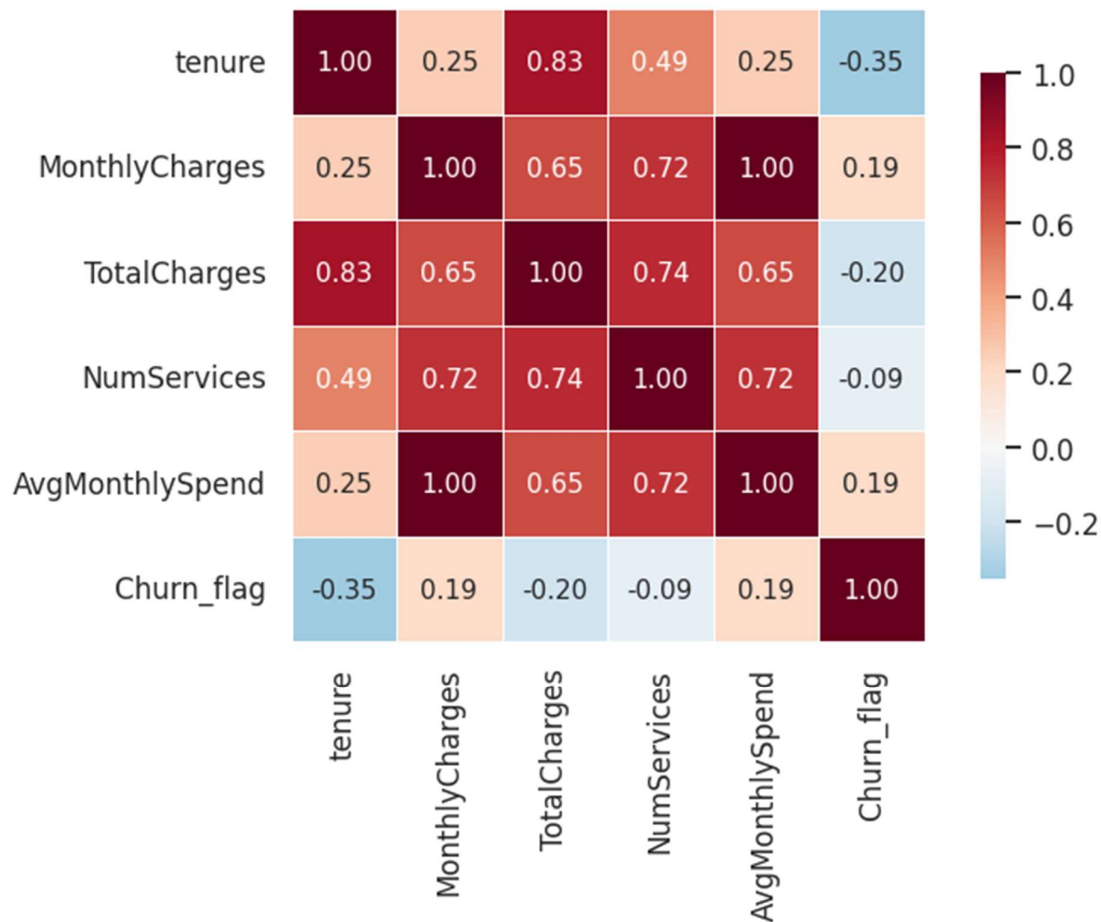


Figure 7. Pearson correlation matrix of the numerical variables

Figure 7 reports three findings relevant to modelling. First, *tenure* has the strongest correlation with *churn* at -0.352 — negative, as expected, and the largest magnitude of any numerical feature. Second, *MonthlyCharges* correlates positively but weakly at $+0.193$, while *TotalCharges* correlates negatively at -0.198 , which is not a contradiction: total charges are largely a function of *tenure*, so the variable inherits *tenure*'s protective association.

Third, and most consequential for the modelling, *tenure* and *TotalCharges* are correlated at 0.826 . This is near-collinearity, and it is structural rather than coincidental, since total charges are approximately *tenure* multiplied by *monthly charge*. Severe multicollinearity inflates the variance of logistic regression coefficient estimates, meaning the coefficients for these two features are individually less reliable even though their joint contribution to prediction is stable. Two consequences follow, both applied in this report: the coefficients in Section 13 are interpreted as indicative of direction and approximate magnitude rather than as precise effect estimates, and a tree-based importance ranking is presented alongside them as an independent cross-check. The condition is recorded in the Limitations section.

Neither feature was dropped. Both carry information relevant to prediction, the objective is predictive accuracy rather than causal coefficient estimation, and the pipeline includes the regularisation that logistic regression applies by default, which moderates the instability.

9. STATISTICAL ANALYSIS

The exploratory analysis identified differences between groups. This section tests formally whether those differences are large enough to be distinguished from sampling variation, and — equally important at a sample size of 7,043 — whether they are large enough to matter.

9.1 Descriptive Statistics

Variable	Count	Mean	Median	Std. dev.	Variance	Min	Max	IQR	Skew
tenure	7,043	32.37	29.00	24.56	603.17	0.00	72.00	46.00	0.24
MonthlyCharges	7,043	64.76	70.35	30.09	905.41	18.25	118.75	54.35	-0.22
TotalCharges	7,043	2279.73	1394.55	2266.79	5138357.17	0.00	8684.80	3388.05	0.96
NumServices	7,043	2.04	2.00	1.85	3.41	0.00	6.00	3.00	0.45

Table 7. Descriptive statistics of the key numerical variables

Table 7 supports several interpretive points. *TotalCharges* is strongly right-skewed (0.96), with a mean of 2,279.73 sitting well above its median of 1,394.55; for this variable the median is the more representative measure of a typical customer, because the mean is pulled upward by the long-tenure tail. Its variance of 5.14 million is likewise dominated by that tail and is the clearest single justification for standardising the numerical features before fitting a linear model.

MonthlyCharges shows mild negative skew (-0.22) with its median (70.35) above its mean (64.76), reflecting the large cluster of low-billing phone-only customers visible in Figure 6. The standard deviation of 24.56 months on tenure, against a mean of 32.37, reflects the bimodality already noted: customers are concentrated at both ends of the lifecycle rather than around its centre.

9.2 Association Between Categorical Variables and Churn

Each categorical variable was tested against churn using the chi-square test of independence, which evaluates the null hypothesis that the variable and churn are independent. Because at $n = 7,043$ even trivial differences achieve significance, Cramér's V is reported alongside each test as a measure of effect size. Cramér's V ranges from 0 to 1, with values around 0.1 conventionally read as weak, 0.3 as moderate and 0.5 as strong.

Variable	Chi-square	p-value	df	Cramér's V	Conclusion
Contract	1184.60	5.86×10^{-258}	2	0.410	Significant — moderate/strong
TenureGroup	856.10	2.94×10^{-185}	3	0.349	Significant — moderate
InternetService	732.31	9.57×10^{-160}	2	0.322	Significant — moderate
PaymentMethod	648.14	3.68×10^{-140}	3	0.303	Significant — moderate
PaperlessBilling	258.28	4.07×10^{-58}	1	0.191	Significant — weak
OnlineSecurity	205.63	1.23×10^{-46}	1	0.171	Significant — weak
Dependents	189.13	4.93×10^{-43}	1	0.164	Significant — weak
TechSupport	190.17	2.92×10^{-43}	1	0.164	Significant — weak
SeniorCitizen	159.43	1.51×10^{-36}	1	0.150	Significant — weak
Partner	158.73	2.14×10^{-36}	1	0.150	Significant — weak
OnlineBackup	47.26	6.21×10^{-12}	1	0.082	Significant — negligible
DeviceProtection	30.51	3.32×10^{-08}	1	0.066	Significant — negligible
StreamingTV	27.86	1.30×10^{-07}	1	0.063	Significant — negligible
StreamingMovies	26.25	3.00×10^{-07}	1	0.061	Significant — negligible
MultipleLines	11.14	0.00084	1	0.040	Significant — negligible
PhoneService	0.92	0.3388	1	0.011	NOT significant

Variable	Chi-square	p-value	df	Cramér's V	Conclusion
gender	0.48	0.4866	1	0.008	NOT significant

Table 8. Chi-square tests of independence with Cramér's V effect sizes ($\alpha = 0.05$)

Fifteen of the seventeen categorical variables are significantly associated with churn at the 5% level. Two are not: gender ($\chi^2 = 0.48$, $p = 0.487$) and PhoneService ($\chi^2 = 0.92$, $p = 0.339$). The gender result deserves emphasis because gender is a conventional segmentation variable in marketing; this dataset provides no evidence that it relates to churn at all, and Section 13 confirms that both the logistic regression and the random forest independently assign it negligible importance. The analysis reached that conclusion from evidence rather than excluding the variable on assumption.

Effect sizes tell the more useful story. Only Contract ($V = 0.410$) reaches a moderate-to-strong association, followed by TenureGroup (0.349), InternetService (0.322) and PaymentMethod (0.303) in the moderate range. Everything below PaperlessBilling is statistically significant but practically weak or negligible. This gap between significance and importance is exactly what a large sample produces, and it is why effect size is reported rather than p -values alone: MultipleLines is "significant" at $p = 0.00084$ while explaining virtually nothing about churn.

9.3 Comparison of Numerical Variables Between Groups

Each numerical variable was compared between churned and retained customers using two tests. Welch's t -test compares means without assuming equal variances, which is appropriate here because the group variances differ. The Mann-Whitney U test is non-parametric and compares distributions without assuming normality, which matters because TotalCharges is skewed and tenure is bimodal. Reporting both establishes that the conclusion does not depend on a distributional assumption. Cohen's d is reported as the standardised effect size, conventionally read as small at 0.2, medium at 0.5 and large at 0.8.

Variable	Mean (churn)	Mean (stay)	Median (churn)	Median (stay)	t-stat	p (Welch)	p (M-W)	Cohen's d
tenure	17.98	37.57	10.00	38.00	-34.82	1.20×10^{-232}	2.42×10^{-208}	-0.893
MonthlyCharges	74.44	61.27	79.65	64.43	18.41	8.59×10^{-73}	3.31×10^{-54}	0.470
TotalCharges	1531.80	2549.91	703.55	1679.52	-18.71	5.90×10^{-75}	5.69×10^{-83}	-0.480
NumServices	1.77	2.14	2.00	2.00	-8.34	9.56×10^{-17}	4.74×10^{-07}	-0.211

Table 9. Welch t -test and Mann-Whitney U comparisons between churned and retained customers

All four variables differ significantly between the groups under both tests, so the findings are robust to the choice of test. The effect sizes rank them clearly. Tenure shows a large effect ($d = -0.893$): churned customers average 17.98 months against 37.57 for retained customers, and the median gap is wider still at 10 against 38 months. MonthlyCharges ($d = 0.470$) and TotalCharges ($d = -0.480$) show medium effects. NumServices shows a small effect ($d = -0.211$), with churners subscribing to 1.77 add-on services on average against 2.14 for retained customers.

One interpretive caution applies throughout this section. Every test reported here establishes association, not causation. The finding that fibre-optic service is associated with higher churn does not establish that fibre causes customers to leave; price, network reliability, competitive intensity in fibre-served areas and the characteristics of customers who choose fibre are all plausible confounders that a single cross-sectional observational dataset cannot separate. The recommendations in Section 17 are framed as interventions to test, not as remedies guaranteed by the evidence.

10. FEATURE ENGINEERING

Three features were derived from the existing variables. Each was motivated by a specific observation from the exploratory analysis, and each uses only information available while the customer is still active, so none introduces leakage.

Feature	Definition	Type	Rationale
TenureGroup	tenure binned into 0–12, 13–24, 25–48 and 49–72 months	Categorical (4 levels)	Figure 3 shows churn risk falls steeply and non-linearly with tenure, and the tenure distribution is bimodal. A linear coefficient on raw tenure cannot represent this; explicit bands let the model learn a distinct risk level for each lifecycle stage. Cramér's V of 0.349 confirms the banded form carries strong signal.
NumServices	Count of the six optional add-on services with value "Yes"	Numerical (0–6)	A proxy for depth of engagement and switching cost. A customer using five services has more to migrate than one using none. The statistical test confirms churners subscribe to fewer add-ons (1.77 against 2.14, $p < 0.001$).
AvgMonthlySpend	TotalCharges ÷ tenure, with MonthlyCharges substituted where tenure = 0	Numerical (continuous)	The customer's historical average monthly bill, which may differ from their current bill after plan changes, promotional expiry or price rises.
ChargeRatio	MonthlyCharges ÷ AvgMonthlySpend	Numerical (continuous)	A price-shock indicator. A ratio above 1 means the current bill exceeds the historical average — a promotional discount has ended or a price rise has landed — which is a plausible trigger for re-evaluating the service. The random forest ranks this seventh of 29 features.

Table 10. Engineered features and their justification

```
# Tenure bands: churn risk is lifecycle-dependent, not linear in months
df['TenureGroup'] = pd.cut(df['tenure'], bins=[-1, 12, 24, 48, 72],
                           labels=['0-12 months', '13-24 months',
                                    '25-48 months', '49-72 months'])

# Engagement depth: how many optional services the customer actually uses
addon_cols = ['OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
              'TechSupport', 'StreamingTV', 'StreamingMovies']
df['NumServices'] = (df[addon_cols] == 'Yes').sum(axis=1)

# Price-shock signal: current bill relative to the historical average
df['AvgMonthlySpend'] = np.where(df['tenure'] > 0,
                                df['TotalCharges'] / df['tenure'],
                                df['MonthlyCharges'])
df['ChargeRatio'] = (df['MonthlyCharges'] /
                    df['AvgMonthlySpend'].replace(0, np.nan)).fillna(1.0)
```

The division in AvgMonthlySpend is guarded against the zero-tenure case, which would otherwise produce infinite values for the eleven new customers identified in Section 6.1; for those customers the current monthly charge is substituted, which is the best available estimate of their spend. ChargeRatio is similarly guarded and defaults to 1.0 where no history exists, representing "no change from baseline". Both guards are necessary precisely because the cleaning stage retained those customers rather than deleting them.

11. MODEL DEVELOPMENT

11.1 Problem Formulation

The target variable *Churn* takes exactly two mutually exclusive values, *Yes* and *No*. The task is therefore binary classification, not regression: the objective is to assign each customer to a class, and the natural output is a probability of class membership rather than a continuous quantity.

Formally, the model estimates $P(\text{Churn} = \text{Yes} \mid X)$, where X is the 23-feature vector described in Table 6, expanding to 29 columns after one-hot encoding. The target was encoded with 1 denoting churn and 0 denoting retention, so that the positive class is the event of interest and precision and recall are computed with respect to churners, which is the operationally meaningful orientation.

This formulation is appropriate because it matches both the data and the decision it supports. A retention team does not need to know how many months remain before a customer leaves; it needs to know whom to contact now. A probability output additionally allows customers to be ranked and a decision threshold to be set according to campaign economics, which is examined in Section 12.4.

11.2 Model Selection

Five classifiers were trained and compared rather than a single model being assumed appropriate. Each was evaluated using stratified five-fold cross-validation on the training partition only, scored on ROC-AUC because that metric is insensitive to class imbalance and to the choice of decision threshold, making it the fairest basis for ranking candidates.

Model	Configuration	Reason for inclusion
Logistic Regression	max_iter = 2000, default class weights	Interpretable linear baseline; coefficients convert directly to odds ratios. Establishes the performance floor.
Logistic Regression (balanced)	max_iter = 2000, class_weight = 'balanced'	Identical model with the loss function reweighted by inverse class frequency, correcting the majority-class bias without resampling.
Decision Tree	max_depth = 6, min_samples_leaf = 25, balanced	Captures non-linear interactions; depth and leaf constraints imposed to limit the overfitting trees are prone to.
Random Forest	n_estimators = 400, max_depth = 12, min_samples_leaf = 5, balanced	Ensemble averaging reduces the variance of a single tree; provides an independent feature importance ranking as a cross-check.
Gradient Boosting	Scikit-learn defaults, random_state = 42	Sequential boosting typically the strongest performer on structured tabular data; included as the accuracy benchmark.

Table 11. Candidate models and the rationale for including each

No resampling technique such as SMOTE or random undersampling was applied. Undersampling would have discarded genuine majority-class observations, and synthetic oversampling would have generated customers who do not exist in order to balance a class ratio that is only moderately skewed. Class weighting achieves the same corrective effect by adjusting the loss function, using only real data, and was preferred on that basis.

11.3 Training the Model

Every model was trained inside a pipeline that encapsulates the preprocessing described in Section 7, so that the scaler and encoder were refitted independently on each cross-validation training fold. The code below is the core of the training loop; the complete listing is given in Appendix A.5.

```
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
results, fitted_models = [], {}

for name, estimator in models.items():
    pipe = Pipeline([('preprocessor', preprocessor), ('classifier', estimator)])
```

```
# Cross-validation on the TRAINING partition only
cv_auc = cross_val_score(pipe, X_train, y_train, cv=cv, scoring='roc_auc')

# Fit on the full training set, then evaluate on the held-out test set
pipe.fit(X_train, y_train)
fitted_models[name] = pipe
y_pred = pipe.predict(X_test)
y_proba = pipe.predict_proba(X_test)[: , 1]
```

The key parameters are as follows. StratifiedKfold with five splits preserves the 26.54% churn rate within every fold, so no fold is evaluated against an unrepresentative class mixture. `max_iter = 2000` on the logistic regressions guarantees convergence of the solver across all 29 features. For the random forest, `max_depth = 12` and `min_samples_leaf = 5` constrain tree growth, and `n_estimators = 400` provides sufficient ensemble averaging to stabilise the result. `random_state = 42` is fixed throughout so that every figure in this report is exactly reproducible from the accompanying notebook.

11.4 Comparative Results and Final Model Choice

Model	CV ROC-AUC	CV std	Accuracy	Precision	Recall	F1-score	Test ROC-AUC
Logistic Regression	0.8459	0.0116	0.7984	0.6520	0.5160	0.5761	0.8425
Logistic Regression (balanced)	0.8457	0.0113	0.7331	0.4983	0.7914	0.6116	0.8423
Decision Tree	0.8234	0.0110	0.7410	0.5076	0.8021	0.6218	0.8345
Random Forest	0.8450	0.0109	0.7722	0.5540	0.7273	0.6289	0.8446
Gradient Boosting	0.8479	0.0124	0.7928	0.6414	0.4973	0.5602	0.8392

Table 12. Cross-validated and test-set performance of the five candidate models

Figure 9. Comparison of five candidate classifiers



Figure 8. Comparison of the five candidate classifiers across evaluation metrics

The five models span only 0.023 in test ROC-AUC, from 0.8345 to 0.8446. This narrow range is itself an informative result: the predictive ceiling is set by the information contained in the available features, not by the sophistication of the algorithm. Gradient boosting, the most complex model, does not outperform a linear model on the ranking metric. Selecting on ROC-AUC alone would therefore be close to arbitrary.

The decision was made instead on the business cost of each error type, which is markedly asymmetric:

- A false negative is a customer who will churn but is predicted safe. No intervention is made and the customer leaves. The cost is the customer's entire remaining lifetime value plus the cost of acquiring a replacement.
- A false positive is a retained customer incorrectly flagged as at risk. They receive a retention offer they did not need. The cost is one discount or one call-centre contact.

In a subscription business the false negative is more expensive by an order of magnitude. Recall on the churn class is therefore the priority metric, with ROC-AUC as the ranking check and generalisation as a constraint.

Final model selected: Logistic Regression with `class_weight = 'balanced'`.

Selection criterion	Result	Assessment
Recall (churn class)	0.7914	Identifies 79.1% of actual churners — second highest of the five, and far above the 0.4973–0.5160 achieved by the unweighted and boosted models
Test ROC-AUC	0.8423	Within 0.0023 of the best model (0.8446); the difference is not meaningful at this sample size
Generalisation gap	0.008 AUC	Train AUC 0.8503 against test AUC 0.8423 — the smallest gap of all five models
Cross-validation stability	std = 0.0113	Consistent performance across all five folds
Interpretability	High	Coefficients convert to odds ratios, so the retention team can see why a customer was flagged, not merely that they were

Table 13. Justification for the final model selection

The two strongest alternatives were rejected for specific reasons. The random forest achieves the highest F1-score (0.6289) and the highest test AUC (0.8446), but its training AUC of 0.9417 against a test AUC of 0.8446 indicates substantial overfitting, and it offers no interpretability advantage to offset that. The unweighted logistic regression reports the more flattering accuracy of 0.7984, but achieves it by identifying only 51.6% of churners — it misses almost half the customers the entire exercise exists to find. This is precisely the failure mode the project guide warns against when accuracy is read in isolation, and it is the clearest demonstration in this study of why metric selection must follow from the problem rather than convention.

12. MODEL EVALUATION

The selected model was evaluated on the held-out test set of 1,409 customers, none of which were seen during training or used in fitting any preprocessing transformation.

12.1 Regression Evaluation

Not applicable. The target variable is categorical with two levels, making this a classification problem. Regression metrics such as MAE, RMSE and R-squared are undefined for a categorical target and are therefore not reported. The evaluation proceeds with classification metrics in Section 12.2.

12.2 Classification Evaluation

Figure 10. Confusion matrix - final model on the held-out test set

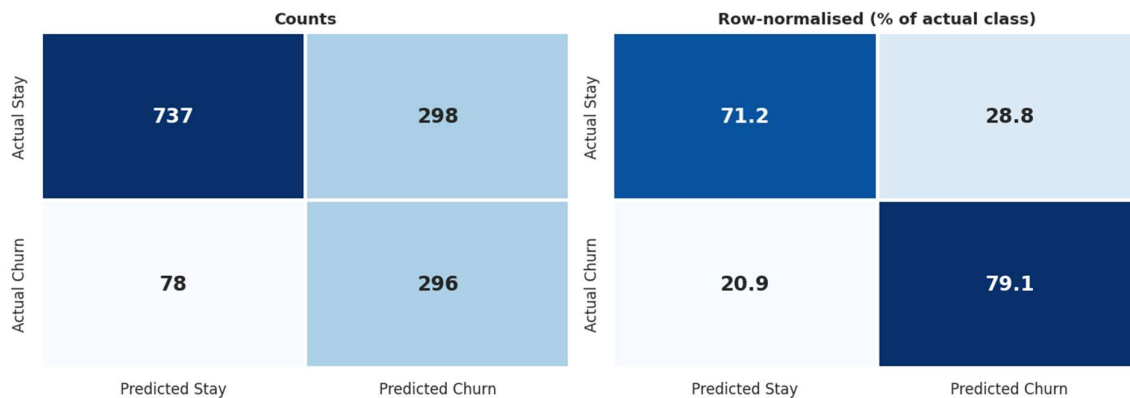


Figure 9. Confusion matrix for the final model on the held-out test set

Outcome	Count	Meaning in this context	Business consequence
True Negative	737	Retained customer correctly predicted to stay	Correctly left alone; no cost incurred
False Positive	298	Retained customer incorrectly flagged as at risk	An unnecessary retention offer — a recoverable, low cost
False Negative	78	Churning customer incorrectly predicted safe	The expensive error: the customer leaves with no intervention
True Positive	296	Churning customer correctly identified	Reachable by a retention campaign before departure

Table 14. Confusion matrix outcomes interpreted in business terms

Metric	Churn class	Retain class	Interpretation
Accuracy	0.7331 (overall)	—	Of 1,409 test customers, 1,033 were classified correctly
Precision	0.4983	0.9043	Of customers flagged as at risk, 49.8% genuinely churned
Recall	0.7914	0.7121	Of customers who actually churned, 79.1% were identified
F1-score	0.6116	0.7968	Harmonic mean of precision and recall for each class
ROC-AUC	0.8423	—	Probability the model ranks a random churner above a random non-churner
PR-AUC	0.6363	—	Average precision across all recall levels, against a 0.2654 base rate
Support	374	1,035	Number of test customers in each class

Table 15. Classification performance of the final model on the test set

Recall of 0.7914 is the headline result: the model identifies approximately four out of every five customers who will churn, while they are still active and therefore still reachable. The 78 false negatives represent 20.9% of actual churners who would leave without any intervention, and they define the practical limit of the model as currently configured.

Precision of 0.4983 means that roughly one in two flagged customers genuinely intends to leave. Read against the 26.54% base rate, this is the more meaningful comparison: a campaign targeting the model's flagged customers achieves a hit rate of 49.8% against 26.5% for a campaign contacting customers at random. The model nearly doubles the efficiency of retention spending. The 298 false positives are the cost of this — at an illustrative retention offer of 500 currency units, approximately 149,000 spent on customers who would have stayed, which is straightforwardly favourable against 296 subscriptions preserved.

Accuracy of 0.7331 is deliberately not the headline. It sits below the 0.7984 achieved by the unweighted model and below the 0.7346 that a classifier predicting "no churn" for every customer would trivially attain. That comparison is the entire argument of Section 11.4: the trivial classifier has an accuracy advantage of 0.0015 over the selected model, and a recall of exactly zero.

Figure 11. Discrimination performance of the classifiers

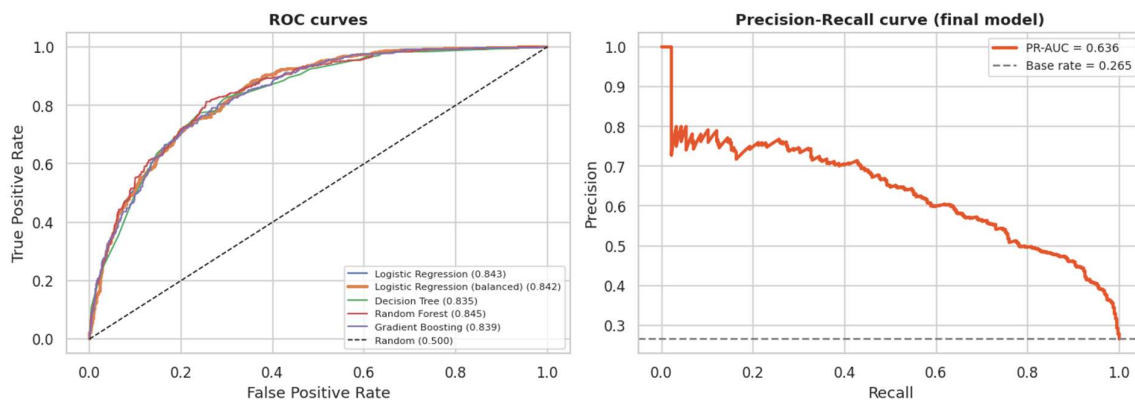


Figure 10. ROC curves for all five models and the precision-recall curve for the final model

The ROC curve in the left panel of Figure 10 plots the true positive rate against the false positive rate across every possible threshold. An AUC of 0.8423 means that given one randomly selected churner and one randomly selected non-churner, the model assigns the churner a higher risk score approximately 84% of the time. All five curves lie well above the diagonal that represents random guessing, confirming genuine predictive signal rather than an artefact of class proportions.

The precision-recall curve in the right panel is the more informative view under class imbalance, because it disregards true negatives, which the majority class supplies in abundance. The curve remains substantially above the 0.2654 base rate across most of the recall range, with a PR-AUC of 0.6363 against a no-skill baseline of 0.2654 — an improvement of roughly 2.4 times. The curve also makes the trade-off explicit: precision degrades gradually up to about 0.7 recall and then falls more sharply, which is the empirical basis for the threshold analysis that follows.

12.3 Overfitting and Underfitting

Generalisation was tested by comparing training and test performance for every candidate model. A model that performs substantially better on data it has seen than on data it has not has memorised rather than learned.

Model	Train accuracy	Test accuracy	Accuracy gap	Train AUC	Test AUC	AUC gap	Assessment
Logistic Regression	0.8094	0.7984	0.0110	0.8503	0.8425	0.0078	Well fitted

Model	Train accuracy	Test accuracy	Accuracy gap	Train AUC	Test AUC	AUC gap	Assessment
Logistic Regression (balanced)	0.7522	0.7331	0.0191	0.8503	0.8423	0.0080	Well fitted
Decision Tree	0.7552	0.7410	0.0142	0.8579	0.8345	0.0234	Slight overfit
Random Forest	0.8562	0.7722	0.0840	0.9417	0.8446	0.0971	Clear overfit
Gradient Boosting	0.8293	0.7928	0.0365	0.8840	0.8392	0.0448	Mild overfit

Table 16. Training versus test performance for all candidate models

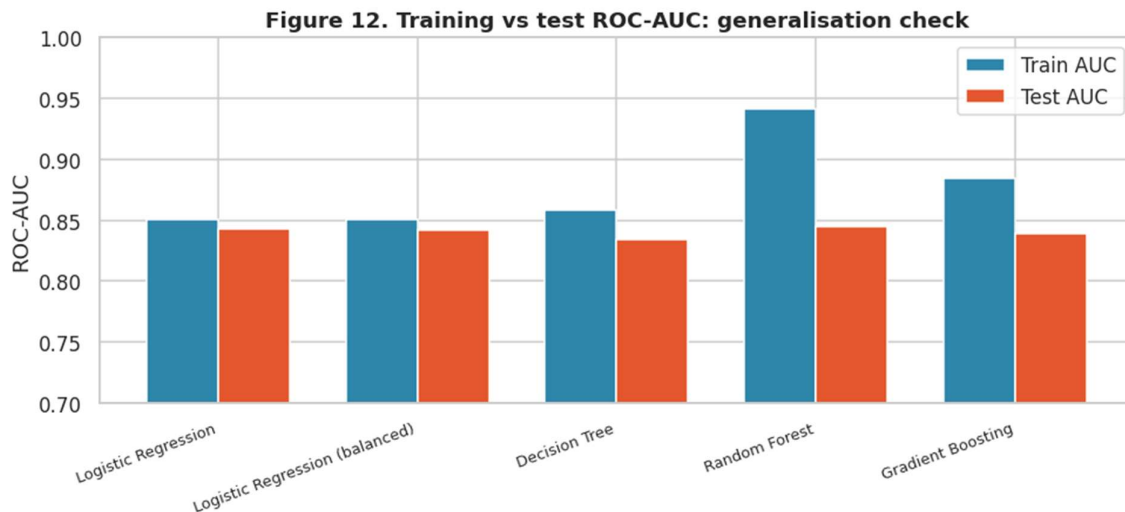


Figure 11. Training versus test ROC-AUC across the five candidate models

Three conclusions follow from Table 16 and Figure 11. The random forest overfits clearly, with an AUC gap of 0.0971: it has learned patterns specific to the training customers that do not transfer to new ones, despite the depth and leaf-size constraints already imposed. Gradient boosting overfits mildly (gap 0.0448) and the decision tree slightly (0.0234). Both logistic regressions generalise cleanly, with gaps of approximately 0.008 — training and test performance are effectively identical, which is the signature of a model whose complexity is well matched to the signal available.

No model underfits. All five achieve test AUC between 0.8345 and 0.8446, far above the 0.50 of random prediction, so none is too simple to capture the structure in the data. In particular, the selected model's combination of strong absolute performance with a near-zero generalisation gap rules out underfitting: a model that was too simple would show poor performance on both partitions, not good performance on both.

This constitutes a second, independent justification for the final model choice. The selection in Section 11.4 was made on the cost asymmetry between error types; the generalisation evidence here supports the same conclusion on entirely separate grounds.

12.4 Decision Threshold Analysis

The default classification threshold of 0.50 is a statistical convention, not a business decision. Because the two error types carry different costs, the threshold was examined explicitly across its operational range.

Threshold	Accuracy	Precision	Recall	F1-score	Missed churners (FN)	Needless offers (FP)	Customers flagged
0.30	0.651	0.427	0.925	0.584	28	464	810

Threshold	Accuracy	Precision	Recall	F1-score	Missed churners (FN)	Needless offers (FP)	Customers flagged
0.35	0.678	0.448	0.906	0.599	35	418	757
0.40	0.705	0.471	0.882	0.614	44	371	701
0.45	0.722	0.486	0.829	0.613	64	328	638
0.50 (default)	0.733	0.498	0.791	0.612	78	298	594
0.55	0.760	0.534	0.751	0.624	93	245	526
0.60	0.774	0.559	0.711	0.626	108	210	476
0.65	0.783	0.581	0.652	0.615	130	176	420

Table 17. Performance across decision thresholds, with error counts in operational terms

The trade-off is clear and monotonic. Lowering the threshold to 0.35 raises recall to 0.906 — only 35 churners missed instead of 78 — at the cost of precision falling to 0.448 and 757 customers being flagged. Raising it to 0.60 improves precision to 0.559 and reduces needless offers to 210, but allows 108 churners to leave undetected.

The operational recommendation follows from the cost of the intervention. Where retention offers are inexpensive and scalable — an email, an in-app credit, an automated message — a threshold of 0.35 to 0.40 is appropriate, because the additional false positives cost little and the recall gain is substantial. Where each intervention is expensive — an outbound call, a device subsidy, a tariff concession — 0.55 to 0.60 better preserves the campaign budget. The default of 0.50 is retained for the headline metrics reported in this section so that results remain directly comparable across the five models.

13. RESULTS AND FINDINGS

This section consolidates the results that answer the research questions set out in Section 4. Feature importance is examined first, since it underpins several of the findings, using two independent methods because the collinearity identified in Section 8.4 makes any single method less reliable on its own.

13.1 Feature Importance

Figure 14. Strongest churn drivers and protectors (logistic regression)

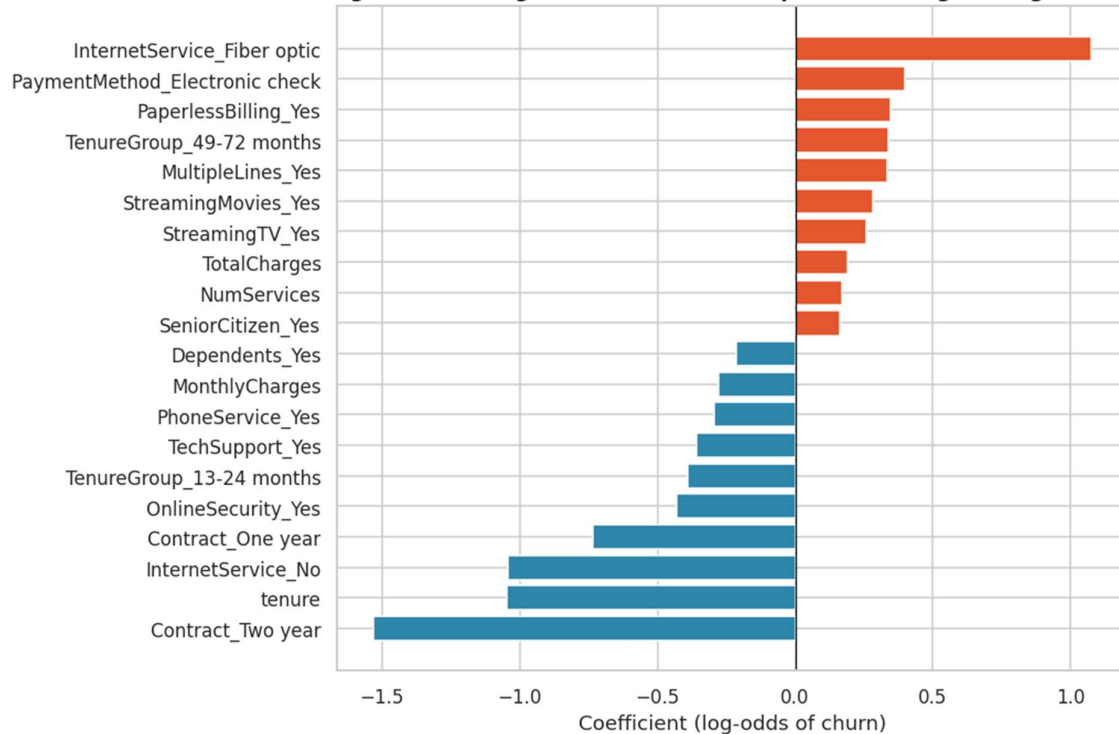


Figure 12. Strongest churn drivers and protective factors (logistic regression coefficients)

Because the numerical features were standardised and the categorical features one-hot encoded, the coefficients are directly comparable. Exponentiating a coefficient converts it to an odds ratio: the multiplicative change in the odds of churn associated with that feature, holding the others constant.

Feature	Coefficient	Odds ratio	Interpretation
Contract — Two year	-1.532	0.216	A two-year contract is associated with roughly 78% lower odds of churn than month-to-month
tenure	-1.044	0.352	Each additional standard deviation of tenure (~24.6 months) is associated with roughly 65% lower odds
InternetService — No	-1.043	0.352	Phone-only customers are the most stable segment in the dataset
Contract — One year	-0.735	0.480	A one-year contract approximately halves the odds relative to month-to-month
OnlineSecurity — Yes	-0.431	0.650	Holding the online security add-on is associated with 35% lower odds
TechSupport — Yes	-0.359	0.698	Holding premium technical support is associated with 30% lower odds
InternetService — Fiber optic	+1.076	2.933	Fibre-optic subscribers face nearly three times the odds of churn, all else equal
PaymentMethod — Electronic check	+0.398	1.489	Manual electronic-check payment is associated with 49% higher odds

Feature	Coefficient	Odds ratio	Interpretation
PaperlessBilling — Yes	+0.347	1.415	Associated with 42% higher odds
MultipleLines — Yes	+0.332	1.394	Higher-value bundles carry greater exposure
StreamingMovies — Yes	+0.281	1.324	Consistent with the bundle-cost effect above

Table 18. Logistic regression coefficients and corresponding odds ratios

The random forest importance ranking, reported in Appendix B, provides an independent cross-check computed by an entirely different mechanism. It ranks tenure highest (0.1459), followed by TotalCharges (0.1021), AvgMonthlySpend (0.0916), MonthlyCharges (0.0896), Contract — Two year (0.0869), InternetService — Fiber optic (0.0741), ChargeRatio (0.0561) and PaymentMethod — Electronic check (0.0550).

The two methods agree on the same small set of drivers: tenure, contract type, fibre-optic service, billing amount and payment method. Agreement between a linear model that estimates coefficients and a tree ensemble that measures impurity reduction provides considerably more confidence than either ranking alone. Notably, gender appears in neither ranking, confirming the chi-square result of $p = 0.487$ through a third independent route.

13.2 Key Findings

The five principal findings of this study are stated below, each with its supporting evidence and practical meaning.

Finding 1 — Contract type is the dominant determinant of churn

Month-to-month customers churn at 42.71%, against 11.27% on one-year contracts and 2.83% on two-year contracts: a fifteen-fold difference between the extremes (Figure 4). The association is statistically significant and has the largest effect size of any categorical variable ($\chi^2 = 1184.60$, $p < 0.001$, Cramér's $V = 0.410$, Table 8), and the two-year contract indicator carries the largest protective coefficient in the model (odds ratio 0.216, Table 18). Because month-to-month is also the largest segment by volume, the highest-risk group is simultaneously the most populous — which makes contract conversion the single highest-leverage intervention available.

Finding 2 — Churn is concentrated in the first year of the customer lifecycle

47.44% of customers in their first twelve months churn, falling to 28.71%, 20.39% and 9.51% in successive tenure bands (Figure 3). Median tenure among churners is 10 months against 38 months among retained customers, and tenure shows the largest effect size of any numerical variable (Cohen's $d = -0.893$, $p < 0.001$, Table 9). Both feature importance methods rank tenure at or near the top. Almost half of all new customers are lost within their first year, which means retention effort concentrated in that window addresses the largest recoverable pool of customers.

Finding 3 — The premium fibre-optic product has the worst retention

Fibre-optic subscribers churn at 41.9%, against 19.0% for DSL and 7.4% for customers without internet service (Figure 5), and fibre carries the largest positive coefficient in the model (odds ratio 2.933, Table 18). The association is significant with a moderate effect size ($\chi^2 = 732.31$, $p < 0.001$, $V = 0.322$). This is a counter-intuitive and commercially serious result: the highest-revenue product loses customers fastest. The dataset contains no service-quality, outage or competitor-pricing variables, so the cause cannot be identified from this analysis — the finding identifies where to investigate, not what to fix.

Finding 4 — Support add-ons are consistently associated with retention

Customers without OnlineSecurity churn at 31.3% against 14.6% for those who hold it, and TechSupport shows a near-identical pattern at 31.2% against 15.2% (Figure 5). Both associations are statistically significant ($p < 0.001$) and both carry protective coefficients in the model (odds ratios 0.650 and 0.698, Table 18). Churners also subscribe to fewer add-on services overall (1.77 against 2.14, $p < 0.001$, Table 9). Whether the add-ons create retention or merely indicate

committed customers cannot be established from observational data, but the consistency of the pattern across two independent add-ons and a composite service count makes it a well-supported target for testing.

Finding 5 — Churn is predictable with operationally useful accuracy

The final model achieves a test ROC-AUC of 0.8423 and identifies 79.14% of churning customers on 1,409 previously unseen records (Tables 12 and 15), with a training-to-test AUC gap of 0.008 confirming that the performance generalises rather than reflecting memorisation (Table 16). Precision of 0.4983 against a base rate of 0.2654 means a targeted campaign is nearly twice as efficient as an untargeted one. The answer to the primary research question is affirmative: churn can be predicted from contract, service and billing attributes with sufficient reliability to support a retention programme.

14. DISCUSSION

The results assemble into a coherent account of why customers leave this operator, and one that is consistent across the exploratory, statistical and modelling stages of the analysis.

The central theme is commitment and friction. The three strongest predictors — contract type, tenure and payment method — all describe how tightly the customer is bound to the relationship rather than what they receive from it. A month-to-month customer paying by electronic check faces a decision point every month: an explicit payment action, no contractual penalty, and an unimpeded exit. A two-year contract holder on automatic payment faces none of these and is not prompted to reconsider at all. The fifteen-fold difference in churn rate between these contractual arrangements is considerably larger than any difference attributable to the services actually delivered. This suggests that churn at this operator is driven less by dissatisfaction with the product than by the absence of anything that sustains the relationship through moments of indifference.

The tenure finding refines this picture. The steep decline in churn across the lifecycle — 47.44% in the first year against 9.51% after four — indicates that the customer relationship passes through a fragile initial phase before stabilising. Customers who survive the first year appear to become progressively more secure, whether through habit, accumulated switching costs, or simply having resolved the problems that arise during onboarding. Two practical implications follow: retention effort has its highest marginal return in the first year, and the volume of customers lost early means the company is continually refilling a base that leaks fastest at the point of entry.

The fibre-optic result runs against the expectation that a premium product commands greater loyalty, and it deserves particular attention because it is the finding most likely to prompt action. Fibre subscribers pay more, receive a superior technical product, and leave at more than twice the rate of DSL customers. The exploratory analysis offers a partial explanation: fibre customers have higher monthly charges, and higher charges are themselves associated with churn. But the effect persists in the multivariate model, where MonthlyCharges is included as a separate feature — fibre carries an odds ratio of 2.933 with price already controlled for. Something specific to the fibre product or to the customers who choose it is therefore at work. Plausible candidates include price-to-value perception at the premium tier, reliability or installation experience, or more aggressive competitor targeting of fibre-eligible addresses. The dataset cannot adjudicate among these, and it would be a misreading of the evidence to present any of them as established.

The protective effect of the support add-ons is open to a similar ambiguity, and it is worth stating the alternative interpretation explicitly rather than adopting the convenient one. The causal reading is that OnlineSecurity and TechSupport create genuine value and switching friction, so providing them reduces churn. The selection reading is that customers who opt into extra services were more committed to begin with, and the add-on is a symptom of that commitment rather than a cause of it. Both are fully consistent with the observed 31% against 15% contrast. The distinction matters commercially: under the causal reading, bundling the add-ons free would reduce churn; under the selection reading, it would not. Only an experiment can separate them, which is why the corresponding recommendation in Section 17 is framed as a controlled trial rather than a rollout.

On the modelling side, the most instructive outcome is how little the choice of algorithm mattered. Five models spanning linear and ensemble approaches fell within 0.023 ROC-AUC of one another, and the most complex model did not lead. The binding constraint is the information available in the features, not the capacity of the algorithm. Adding gradient boosting to a problem of this kind buys very little; adding a service-quality variable or a complaint history would likely buy a great deal. This reframes where effort should go in any follow-up work — toward data acquisition rather than model tuning.

Equally instructive is the divergence between accuracy and usefulness. The unweighted logistic regression is the more accurate model by 6.5 percentage points and the less useful one by a wide margin, because it purchases that accuracy by failing to identify nearly half of all churners. A trivial classifier predicting "no churn" universally would be more accurate still, at 73.46%, and entirely worthless. The selected model is less accurate than both and is the correct choice,

because in a problem with asymmetric error costs and an imbalanced target, accuracy measures the wrong thing. The model should be judged against the decision it supports, not against a metric chosen by convention.

A final observation concerns what a practitioner should take from the predictions themselves. At the default threshold the model flags 594 of 1,409 customers and is correct about half the time. It is not an oracle and should not be deployed as one. Its value is that it raises the hit rate of retention activity from 26.5% to 49.8% and supplies a ranked probability that allows finite outreach capacity to be allocated where it is most likely to matter. That is a substantial operational improvement, and framing it honestly is more useful than presenting the model as a solution to churn.

15. LIMITATIONS

The following limitations constrain the conclusions of this study and should be weighed by anyone acting on them.

Data limitations

- Observational design, no causal inference. The dataset records what customers did, not what would have happened under a different treatment. Every relationship reported is an association. The fibre-optic and payment-method findings in particular are open to confounding by variables not present in the data, and no recommendation in Section 17 should be read as a guaranteed remedy.
- Single cross-sectional snapshot with no time dimension. There are no timestamps, no record of when each customer joined in calendar terms, and no indication of the period the "last month" represents. Seasonal effects, the impact of specific price changes or competitor launches, and any trend in churn over time are therefore invisible. Survival analysis, which would model when a customer churns rather than whether, is not possible with this structure.
- Important variables are absent. The dataset contains no service-quality or outage records, no complaint or support-ticket history, no customer satisfaction or NPS score, no competitor pricing, and no record of previous retention offers made. These are among the most plausible drivers of churn, and their omission means the model is explaining churn using proxies for causes it cannot observe — which is the most likely reason all five algorithms plateaued near 0.84 ROC-AUC.
- Single-operator, single-market data. All 7,043 customers belong to one company. Market structure, regulatory environment, pricing norms and competitive intensity differ across operators and countries, so the coefficients reported here should not be transferred to another business without revalidation.
- The dataset is a widely distributed educational sample of a fictional operator. While realistic in structure, it may not reproduce the full messiness, rare cases and measurement error of production customer data.

Methodological and modelling limitations

- Multicollinearity between tenure and TotalCharges ($r = 0.826$) inflates the variance of the logistic regression coefficient estimates. The individual coefficients for these two features are therefore less reliable than their magnitudes suggest, although their joint contribution to prediction is stable. This is why the tree-based importance ranking is presented alongside them as a cross-check.
- Performance is validated on a single 80/20 split with a fixed random seed. Although five-fold cross-validation was used for model selection and produced a low standard deviation (0.0113), repeated splitting or nested cross-validation would give a more robust estimate of the final test metrics.
- Hyperparameters were set from reasonable defaults and standard practice rather than systematic search. A grid or randomised search, properly nested inside the cross-validation to avoid selection bias, might improve performance modestly — though the narrow spread across five very different algorithms suggests the gain would be small.
- The model is static. It would degrade as pricing, product mix and competitive conditions change, and would require periodic retraining and monitoring in any real deployment. No drift monitoring is addressed here.
- 20.9% of churners are missed at the default threshold. The model reduces but does not eliminate undetected churn, and the threshold analysis in Section 12.4 shows that improving this necessarily costs precision.
- The cost-benefit illustration in Section 12.2 uses an assumed retention offer value of 500 currency units. Actual figures for customer lifetime value, offer cost and offer success rate were not available, so the economic argument is directional rather than quantified.

Scope limitations

- Only classification was attempted. Customer lifetime value modelling, uplift modelling (which predicts who will respond to an intervention rather than who will churn) and time-to-churn modelling are all outside the scope of this study, though each would strengthen a retention programme.
- Fairness across demographic groups was not formally audited. Although gender was found to be statistically unrelated to churn, SeniorCitizen carries a significant association, and any deployment of this model should be assessed for disparate impact before it is used to allocate offers.

16. CONCLUSION

This project set out to determine whether subscription customer churn can be predicted from contract, service and billing attributes while customers are still active, and to identify which attributes distinguish those who leave. Both objectives were met.

A dataset of 7,043 telecommunications customers was assessed, cleaned and analysed without the loss of a single observation. The three data quality issues identified — eleven blank TotalCharges entries, redundant labels across seven service columns, and one mis-typed variable — were each diagnosed before treatment, and the treatment chosen in each case followed from what the diagnosis revealed rather than from convenience. The eleven blank entries were found to belong exclusively to customers with zero tenure, making zero the factually correct value and deletion or median imputation both incorrect.

The analysis established that churn in this customer base is overwhelmingly a function of commitment structure rather than service consumption. Contract type is the strongest single determinant, with month-to-month customers churning at 42.71% against 2.83% on two-year contracts. Churn is heavily concentrated in the first year of the relationship, at 47.44%, falling to 9.51% beyond four years. Fibre-optic service, electronic-check payment and the absence of support add-ons are each independently associated with elevated risk, while gender — a conventional segmentation variable — shows no significant association at all. Each of these conclusions was verified through formal statistical testing with effect sizes reported alongside significance, and each is corroborated by the feature importance of the fitted model.

Five classification models were compared. A class-weighted logistic regression was selected, achieving 0.8423 ROC-AUC and 0.7914 recall on 1,409 held-out customers, with a training-to-test AUC gap of 0.008 confirming genuine generalisation. The selection was made on the asymmetry between error costs rather than on accuracy, and the model chosen is deliberately less accurate than two alternatives that identify only half as many churners. The narrow performance spread across five structurally different algorithms indicates that the limiting factor is the information available in the data, not the sophistication of the method.

The practical contribution is a model that identifies approximately four in five departing customers while they can still be reached, and raises the hit rate of retention activity from 26.5% to 49.8%. Its limitations are equally clear: it is built on observational data from a single operator, it misses a fifth of churners at the default threshold, and it identifies where to investigate rather than what to fix. Within those boundaries, the project demonstrates that a disciplined analytical pipeline — diagnosis before treatment, statistical verification of visual findings, metric selection driven by business cost, and explicit testing for overfitting — produces conclusions that are both defensible and actionable.

17. RECOMMENDATIONS AND FUTURE WORK

17.1 Recommendations

The five recommendations below follow the structure finding → evidence → implication → action. Each is traceable to a specific result in this report.

Recommendation 1 — Establish a structured first-year onboarding and retention programme

- Finding: churn is concentrated in the first twelve months, at 47.44%, falling to 9.51% beyond four years (Figure 3).
- Evidence: tenure carries the largest numerical effect size in the study (Cohen's $d = -0.893$, $p < 0.001$) and is ranked first by the random forest importance measure.
- Implication: almost half of all newly acquired customers are lost before the relationship stabilises, so acquisition spending is being partially wasted at the point of entry.
- Action: implement structured touchpoints at months 1, 3 and 6 — an onboarding check that the service is working as expected, usage guidance addressing features the customer has not activated, and a milestone incentive at month 6. Measure the programme against a holdout control group so its effect can be isolated.

Recommendation 2 — Prioritise conversion of month-to-month customers to annual contracts

- Finding: month-to-month customers churn at 42.71% against 11.27% on one-year and 2.83% on two-year contracts (Figure 4).
- Evidence: the largest effect size of any categorical variable (Cramér's $V = 0.410$) and the largest protective coefficient in the model (odds ratio 0.216 for two-year contracts).
- Implication: the step from month-to-month to one year produces a far larger reduction in risk than the step from one to two years, so the first conversion carries most of the value.
- Action: offer a discount worth less than one month of revenue in exchange for a twelve-month commitment, targeting month-to-month customers whose model risk score exceeds the operating threshold. The economics are favourable whenever the discount costs less than one month of margin and the conversion prevents a departure worth eleven.

Recommendation 3 — Commission an investigation into fibre-optic customer experience

- Finding: fibre-optic subscribers churn at 41.9% against 19.0% for DSL, and retain the highest positive coefficient in the model even with monthly charges controlled for (odds ratio 2.933).
- Evidence: $\chi^2 = 732.31$, $p < 0.001$, Cramér's $V = 0.322$; the effect persists in the multivariate model.
- Implication: the highest-revenue product has the worst retention, which represents the largest single revenue exposure identified in this analysis. The dataset cannot establish why.
- Action: gather the data required to diagnose it — outage and service-quality logs, installation experience scores, support ticket volumes and competitor pricing in fibre-served areas — and conduct customer interviews with recent fibre churners. This is a diagnostic exercise, not a remediation, and should be resourced as such.

Recommendation 4 — Trial free OnlineSecurity and TechSupport for new customers

- Finding: customers without OnlineSecurity churn at 31.3% against 14.6% with it; TechSupport shows 31.2% against 15.2% (Figure 5).
- Evidence: both associations significant at $p < 0.001$, both carrying protective coefficients (odds ratios 0.650 and 0.698); churners also hold fewer add-ons overall (1.77 against 2.14).
- Implication: the direction of causation is genuinely uncertain — the add-ons may create retention, or committed customers may simply buy more — and the two readings imply different returns on the same intervention.

- Action: run a randomised trial rather than a rollout. Offer both add-ons free for six months to a random half of new customers and compare twelve-month retention against the control. This costs little, resolves the causal ambiguity directly, and converts a retention lever into a cross-sell opportunity if it succeeds.

Recommendation 5 — Migrate electronic-check customers to automatic payment

- Finding: electronic-check payers churn at 45.3% against 16.7% on automatic bank transfer and 15.2% on credit card (Figure 6).
- Evidence: $\chi^2 = 648.14$, $p < 0.001$, Cramér's $V = 0.303$; electronic check carries an odds ratio of 1.489 in the model.
- Implication: a manual payment creates a recurring decision point at which the customer re-evaluates the service. Removing it removes the prompt — though it is also possible that less committed customers simply prefer manual payment.
- Action: offer a one-time bill credit for switching to automatic payment, targeted at electronic-check customers. This is inexpensive and operationally simple; as with Recommendation 4, it should be measured against a control group because the causal direction is not established.

17.2 Future Work

Four extensions would materially strengthen this analysis, listed in order of expected return.

7. Acquire the missing explanatory variables. The consistent plateau of all five models near 0.84 ROC-AUC indicates a ceiling imposed by the available features, not by the algorithms. Service-quality records, complaint and support history, satisfaction scores, competitor pricing and prior retention contacts are the most promising additions, and would likely yield a greater improvement than any amount of model tuning.
8. Move from cross-sectional to longitudinal data. With timestamps and repeated observations of the same customers, survival analysis (Cox proportional hazards or an accelerated failure time model) would predict when a customer is likely to churn rather than only whether, allowing interventions to be timed rather than merely targeted.
9. Develop uplift modelling. This study predicts who will churn; a retention programme actually needs to know who will respond to an intervention. Some at-risk customers will leave regardless, and some would have stayed without contact. Uplift modelling, built on data from randomised retention trials, targets the persuadable segment and is the natural successor to this work.
10. Strengthen validation and deployment practice. Repeated or nested cross-validation would give a more robust performance estimate; nested hyperparameter search would tune without selection bias; a fairness audit across demographic groups should precede any live use; and a deployed model would require drift monitoring and a scheduled retraining cycle as pricing and competitive conditions evolve.

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APPENDIX A. CODE

The complete, executable analysis is contained in the accompanying notebook, *IDRA_Capstone_Customer_Churn.ipynb*, which runs from start to finish in order. The listings below reproduce the code essential to reproducing each stage.

A.1 Data Loading and Inspection

```
import numpy as np, pandas as pd
import matplotlib.pyplot as plt, seaborn as sns
from scipy import stats

RANDOM_STATE = 42
np.random.seed(RANDOM_STATE)

df = pd.read_csv('Customer-Churn_.csv')
print(f'Rows: {df.shape[0]:,} Columns: {df.shape[1]:,}') # 7,043 x 21
df.info()
df.describe().T

print(df['Churn'].value_counts(normalize=True) * 100) # 73.46 / 26.54
```

A.2 Data Cleaning

```
# --- Hidden missing values -----
blank_mask = df['TotalCharges'].astype(str).str.strip() == ''
print(blank_mask.sum(), df.loc[blank_mask, 'tenure'].unique()) # 11, [0]

df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
df.loc[df['TotalCharges'].isna(), 'TotalCharges'] = 0.0

# --- Duplicates -----
print(df.duplicated().sum(), df['customerID'].duplicated().sum()) # 0, 0

# --- Inconsistent category labels -----
service_cols = ['MultipleLines', 'OnlineSecurity', 'OnlineBackup',
                'DeviceProtection', 'TechSupport', 'StreamingTV',
                'StreamingMovies']
for col in service_cols:
    df[col] = df[col].replace({'No internet service': 'No',
                              'No phone service': 'No'})

# --- Data types -----
df['SeniorCitizen'] = df['SeniorCitizen'].map({0: 'No', 1: 'Yes'})

# --- Outlier assessment (IQR rule) -----
for col in ['tenure', 'MonthlyCharges', 'TotalCharges']:
    q1, q3 = df[col].quantile([0.25, 0.75]); iqr = q3 - q1
    lo, hi = q1 - 1.5 * iqr, q3 + 1.5 * iqr
    print(col, ((df[col] < lo) | (df[col] > hi)).sum()) # 0 in every case
```

A.3 Feature Engineering and Preprocessing

```
# --- Engineered features -----
df['TenureGroup'] = pd.cut(df['tenure'], bins=[-1, 12, 24, 48, 72],
                           labels=['0-12 months', '13-24 months',
                                   '25-48 months', '49-72 months'])

addon_cols = ['OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
              'TechSupport', 'StreamingTV', 'StreamingMovies']
df['NumServices'] = (df[addon_cols] == 'Yes').sum(axis=1)

df['AvgMonthlySpend'] = np.where(df['tenure'] > 0,
                                df['TotalCharges'] / df['tenure'],
                                df['MonthlyCharges'])
df['ChargeRatio'] = (df['MonthlyCharges'] /
                    df['AvgMonthlySpend']).replace(0, np.nan).fillna(1.0)
df['Churn_flag'] = (df['Churn'] == 'Yes').astype(int)

# --- Feature / target separation -----
X = df.drop(columns=['customerID', 'Churn', 'Churn_flag'])
y = df['Churn_flag']
X['TenureGroup'] = X['TenureGroup'].astype(str)

numeric_features = X.select_dtypes(include=np.number).columns.tolist()
categorical_features = [c for c in X.columns if c not in numeric_features]

# --- Split BEFORE fitting any transformation -----
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=RANDOM_STATE, stratify=y)

preprocessor = ColumnTransformer(transformers=[
    ('num', StandardScaler(), numeric_features),
    ('cat', OneHotEncoder(drop='first', handle_unknown='ignore'),
     categorical_features)])
```

A.4 Statistical Testing

```
def cramers_v(confusion):
    chi2 = stats.chi2_contingency(confusion)[0]
    n = confusion.to_numpy().sum()
    r, k = confusion.shape
    return np.sqrt((chi2 / n) / (min(r - 1, k - 1)))

# Categorical variables: chi-square with effect size
for col in cat_cols:
    ct = pd.crosstab(df[col], df['Churn'])
    chi2, p, dof, _ = stats.chi2_contingency(ct)
    print(col, round(chi2, 2), p, round(cramers_v(ct), 3))

# Numerical variables: Welch t-test, Mann-Whitney U and Cohen's d
for col in ['tenure', 'MonthlyCharges', 'TotalCharges', 'NumServices']:
    a = df.loc[df.Churn_flag == 1, col]
    b = df.loc[df.Churn_flag == 0, col]
    t, p_t = stats.ttest_ind(a, b, equal_var=False)
    u, p_u = stats.mannwhitneyu(a, b, alternative='two-sided')
    d = (a.mean() - b.mean()) / np.sqrt((a.var(ddof=1) + b.var(ddof=1)) / 2)
```

APPENDIX B. ADDITIONAL FIGURES AND TABLES

The material below supports the main analysis but was not essential to its narrative.

B.1 Demographic Variables and Churn

Figure 7. Demographic variables: only senior-citizen status shows a marked gap

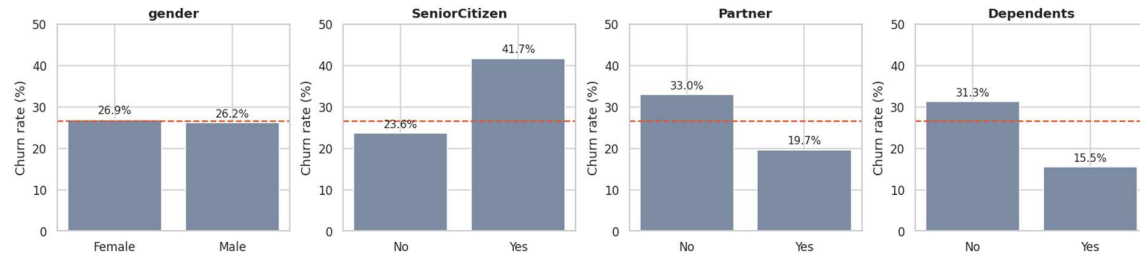


Figure 13. Churn rate across the four demographic variables

Gender shows effectively no difference (26.9% against 26.2%), consistent with the non-significant chi-square result reported in Table 8. Senior citizens churn at 41.7% against 23.6% for non-seniors, and customers without a partner or without dependents churn more than those with. These variables were retained in the feature set so that the model could assess them on evidence rather than have them excluded by assumption; both feature importance methods subsequently ranked gender near zero.

B.2 Decision Threshold Trade-off

Figure 13. Precision-recall trade-off across decision thresholds

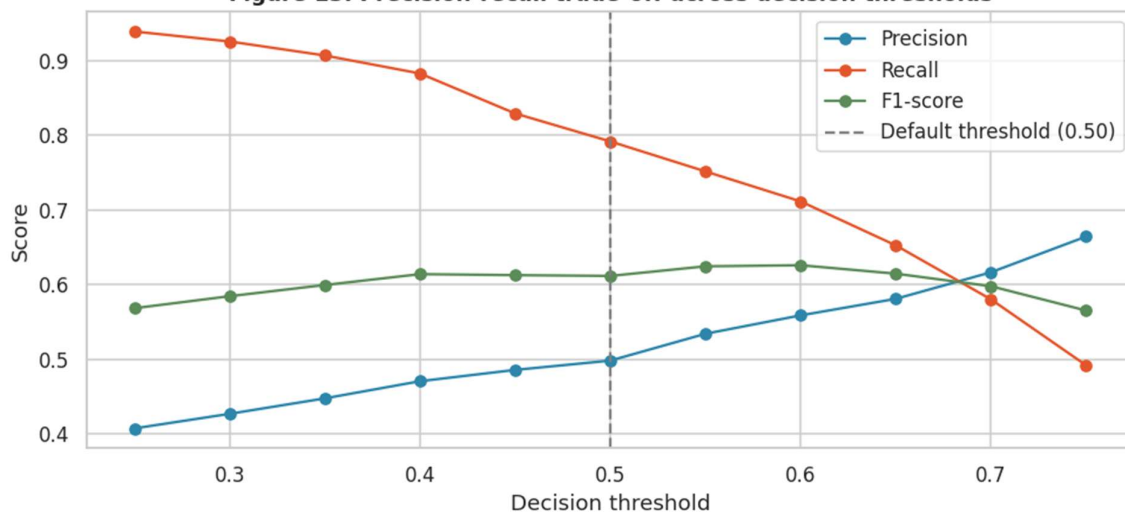


Figure 14. Precision, recall and F1-score across decision thresholds

This figure is the graphical form of Table 17. The crossing point of the precision and recall curves lies above the default threshold of 0.50, which reflects the class weighting applied during training. F1-score is comparatively flat between 0.40 and 0.65, indicating that the threshold can be tuned to the economics of the retention campaign across a wide range without materially degrading balanced performance.

B.3 Random Forest Feature Importance

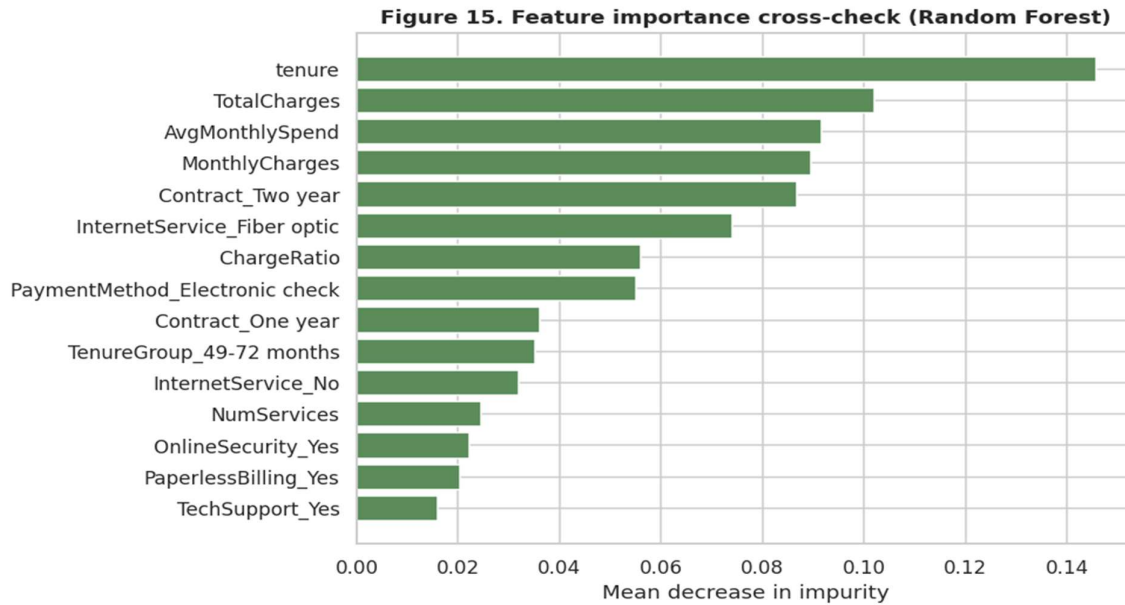


Figure 15. Feature importance from the random forest (mean decrease in impurity)

This ranking is the independent cross-check referenced in Section 13.1. It is computed by an entirely different mechanism from the logistic regression coefficients — measuring how much each feature reduces impurity across 400 trees — yet identifies the same set of dominant drivers, which materially strengthens confidence in those findings.

Rank	Feature	Importance	Rank	Feature	Importance
1	tenure	0.1459	7	ChargeRatio	0.0561
2	TotalCharges	0.1021	8	PaymentMethod — Electronic check	0.0550
3	AvgMonthlySpend	0.0916	9	Contract — One year	0.0362
4	MonthlyCharges	0.0896	10	TenureGroup — 49-72 months	0.0351
5	Contract — Two year	0.0869	11	InternetService — No	0.0320
6	InternetService — Fiber optic	0.0741	12	NumServices	0.0246

Table 19. Top twelve features by random forest importance

B.4 Risk Band Segmentation

Converting the model's probability output into risk bands turns it into an operational tool, computed here on the 1,409 test customers.

Risk band	Probability range	Customers	Actual churners	Actual churn rate	Share of all churners
Low	0.00 – 0.35	652	35	5.4%	9.4%
Medium	0.35 – 0.65	337	95	28.2%	25.4%
High	0.65 – 1.00	420	244	58.1%	65.2%

Table 20. Customer risk bands derived from the model probability output

End of report.